LMC6482 CMOS Rail-to-Rail Input and Output Operational Amplifier

1 Features

- Rail-to-rail input common-mode voltage range (specified over temperature)
- Rail-to-rail output swing (within 20-mV of supply rail, 100-kΩ load)
- Specified 3-V, 5-V, and 15-V performance
- Excellent CMRR and PSRR: 82 dB
- Ultra-low input current: 20 fA
- Specified for 2-kΩ and 600-Ω loads
- Improved replacement for TLC272, TLC277

2 Applications

- Data acquisition (DAQ)
- Currency counter
- Oscilloscope (DSO)
- Intra-DC interconnect (METRO)
- Macro remote radio unit (RRU)
- Multiparameter patient monitor
- Merchant telecom rectifiers
- Train control and management
- Process analytics (pH, gas, concentration, force, and humidity)
- Three phase UPS

3 Description

The LMC6482 provides a common-mode range that extends to both supply rails. This rail-to-rail performance combined with excellent accuracy, due to a high CMRR, makes these devices unique among rail-to-rail input amplifiers. The device is an excellent choice for systems, such as data acquisition, that require a large input signal range. The LMC6482 is also an excellent upgrade for circuits using limited common-mode range amplifiers, such as the TLC272 and TLC277.

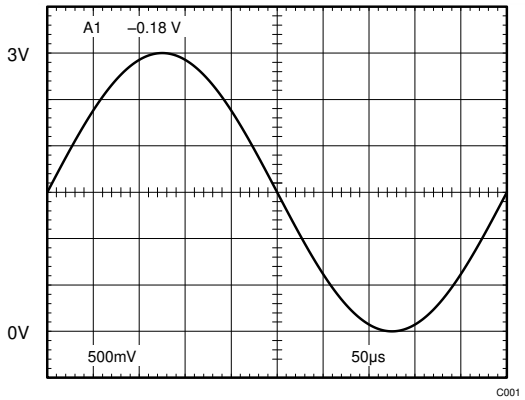
Maximum dynamic signal range is provided in low voltage and single supply systems by the rail-to-rail output swing of the LMC6482. The rail-to-rail output swing is maintained for loads down to 600 Ω of the device. Specified low-voltage characteristics and low-power dissipation make the LMC6482 a great choice for battery-operated systems.

The LMC6482 is available in PDIP, SOIC, and VSSOP packages.

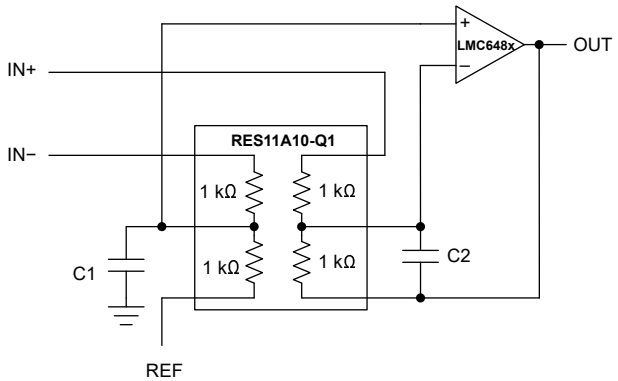
Device Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)
LMC6482	D (SOIC, 8)	4.9 mm × 6 mm
	DGK (VSSOP, 8)	3 mm × 4.9 mm
	P (PDIP, 8)	9.81 mm × 9.43 mm

(1) For more information, see Section 10.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Rail-to-Rail Input (V_S = 3 V)



Unity-Gain Difference Amplifier

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Pin Configuration and Functions

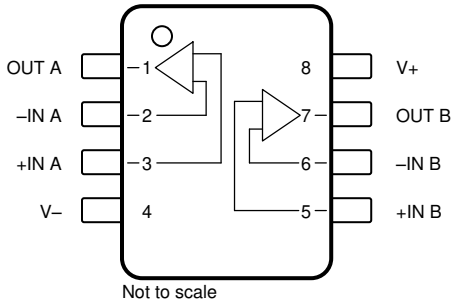


Figure 4-1. D, DGK, and P Packages, 8-Pin SOIC, VSSOP, and PDIP (Top View)

Table 4-1. Pin Functions

PIN		TYPE	DESCRIPTION
NO.	NAME		
1	OUT A	Output	Output for amplifier A
2	−IN A	Input	Inverting input for amplifier A
3	+IN A	Input	Noninverting input for amplifier A
4	V–	Power	Negative supply voltage input
5	+IN B	Input	Noninverting input for amplifier B
6	−IN B	Input	Inverting input for amplifier B
7	OUT B	Output	Output for amplifier B
8	V+	Power	Positive supply voltage input

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5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1) (2)

		MIN	MAX	UNIT
	Differential input voltage		±Supply Voltage	
	Voltage at input/output pin	(V–) – 0.3	(V+) + 0.3	V
V_S	Supply voltage, V_S = (V+) – (V–)		16	V
	Current at input pin(3)	–5	5	mA
	Current at output pin(4) (5)	–30	30	mA
	Current at power supply pin		40	mA
	Lead temperature (soldering, 10 sec)		260	°C
T_J	Junction temperature(6)		150	°C
T_{STG}	Storage temperature	–65	150	°C

(1) Operation outside the Absolute Maximum Ratings may cause permanent device damage. Absolute Maximum Ratings do not imply functional operation of the device at these or any other conditions beyond those listed under Recommended Operating Conditions. If used outside the Recommended Operating Conditions but within the Absolute Maximum Ratings, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
(2) If military- or aerospace-specified devices are required, contact the TI Sales Office or Distributors for availability and specifications.
(3) Limiting input pin current is only necessary for input voltages that exceed absolute maximum input voltage ratings.
(4) Applies to both single-supply and split-supply operation. Continuous short circuit operation at elevated ambient temperature can result in exceeding the maximum allowed junction temperature of 150°C. Output currents in excess of ±30 mA over a long term can adversely affect reliability.
(5) Do not short circuit output to V+, when V+ is greater than 13 V or reliability is adversely affected.
(6) The maximum power dissipation is a function of $T_{J(max)}$, $R_{\theta JA}$, and T_A. The maximum allowable power dissipation at any ambient temperature is $P_D = (T_{J(max)} - T_A) / \theta_{JA}$. All numbers apply for packages soldered directly into a printed circuit board (PCB).

5.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±1500	V

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
V_S	Supply voltage, V_S = (V+) – (V–)	3		15.5	V
T_J	Junction temperature	–40		85	°C

5.4 Thermal Information

THERMAL METRIC(1)		LMC6482			UNIT
		D (SOIC)	DGK (VSSOP)	P (PDIP)	
		8 PINS	8 PINS	8 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	155	194	90	°C/W

(1) For more information about traditional and new thermal metrics, see the Semiconductor and IC Package Thermal Metrics application report.

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5.5 Electrical Characteristics: V_S = 5 V

at T_J = +25°C, V+ = 5 V, V– = 0 V, V_{CM} = V_{OUT} = V+ / 2, and R_L > 1 MΩ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
DC SPECS							
V_{OS}	Input offset voltage	LMC6482AI			±0.11	±0.75	mV
			T_A = –40°C to +85°C			±1.35	
		LMC6482I			±0.11	±3	
			T_A = –40°C to +85°C			±3.7	
dV_{OS}/dT	Input offset voltage drift	T_A = –40°C to +85°C			±1		µV/°C
I_B	Input bias current				±0.02		pA
		T_A = –40°C to +85°C				±4	
I_{OS}	Input offset current				±0.01		pA
		T_A = –40°C to +85°C				±2	
C_{IN}	Common-mode input capacitance				3		pF
R_{IN}	Input resistance				10		TΩ
CMRR	Common-mode rejection ratio	LMC6482AI 0 V < V_{CM} < 15 V, V+ = 15 V		70	82		dB
			T_A = –40°C to +85°C	67			
		LMC6482I 0 V < V_{CM} < 15 V, V+ = 15 V		65	82		
			T_A = –40°C to +85°C	62			
		LMC6482AI 0 V < V_{CM} < 5 V, V+ = 5 V		70	82		
			T_A = –40°C to +85°C	67			
		LMC6482I 0 V < V_{CM} < 5 V, V+ = 5 V		65	82		
			T_A = –40°C to +85°C	62			
+PSRR	Positive power-supply rejection ratio	LMC6482AI 5 V < V+ < 15 V, V– = 0 V, V_O = 2.5 V		70	82		dB
			T_A = –40°C to +85°C	67			
		LMC6482I 5 V < V+ < 15 V, V– = 0 V, V_O = 2.5 V		65	82		
			T_A = –40°C to +85°C	62			
–PSRR	Negative power-supply rejection ratio	LMC6482AI –5 V < V– < –15 V, V+ = 0 V, V_O = –2.5 V		70	82		dB
			T_A = –40°C to +85°C	67			
		LMC6482I –5 V < V– < –15 V, V+ = 0 V, V_O = –2.5 V		65	82		
			T_A = –40°C to +85°C	62			
V_{CM}	Input common-mode voltage	V+ = 5 V and 15 V, for CMRR ≥ 50 dB	Low		(V–) – 0.3	–0.25	V
			Low, T_A = –40°C to +85°C			0	
			High	(V+) + 0.25	(V+) + 0.3		
			High, T_A = –40°C to +85°C	(V+)			

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5.5 Electrical Characteristics: V_S = 5 V (continued)

at T_J = +25°C, V+ = 5 V, V– = 0 V, V_{CM} = V_{OUT} = V+ / 2, and R_L > 1 MΩ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
A_V	Large-signal voltage gain	LMC6482AI sourcing, R_L = 2 kΩ to 7.5 V, V+ = 15 V, 7.5 V ≤ V_O ≤ 11.5 V		140	666		V/mV
			T_A = –40°C to +85°C	84			
		LMC6482I sourcing, R_L = 2 kΩ to 7.5 V, V+ = 15 V, 7.5 V ≤ V_O ≤ 11.5 V		120	666		
			T_A = –40°C to +85°C	72			
		LMC6482AI sinking, R_L = 2 kΩ to 7.5 V, V+ = 15 V, 3.5 V ≤ V_O ≤ 7.5 V		35	75		
			T_A = –40°C to +85°C	20			
		LMC6482I sinking, R_L = 2 kΩ to 7.5 V, V+ = 15 V, 3.5 V ≤ V_O ≤ 7.5 V		35	75		
			T_A = –40°C to +85°C	20			
		LMC6482AI sourcing, R_L = 600 Ω to 7.5 V, V+ = 15 V, 7.5 V ≤ V_O ≤ 11.5 V		80	300		
			T_A = –40°C to +85°C	48			
		LMC6482I sourcing, R_L = 600 Ω to 7.5 V, V+ = 15 V, 7.5 V ≤ V_O ≤ 11.5 V		50	300		
			T_A = –40°C to +85°C	30			
		LMC6482AI sinking, R_L = 600 Ω to 7.5 V, V+ = 15 V, 3.5 V ≤ V_O ≤ 7.5 V		20	35		
			T_A = –40°C to +85°C	13			
		LMC6482I sinking, R_L = 600 Ω to 7.5 V, V+ = 15 V, 3.5 V ≤ V_O ≤ 7.5 V		15	35		
			T_A = –40°C to +85°C	10			
V_O	Voltage output swing	V+ = 5 V, R_L = 2 kΩ to V+ / 2	Swing high	4.8	4.9		V
			Swing high, T_A = –40°C to +85°C	4.7			
			Swing low		0.1	0.18	
			Swing low, T_A = –40°C to +85°C			0.24	
		V+ = 5 V, R_L = 600 Ω to V+ / 2	Swing high	4.5	4.7		
			Swing high, T_A = –40°C to +85°C	4.24			
			Swing low		0.3	0.5	
			Swing low, T_A = –40°C to +85°C			0.65	
		V+ = 15 V, R_L = 2 kΩ to V+ / 2	Swing high	14.4	14.7		
			Swing high, T_A = –40°C to +85°C	14.2			
			Swing low		0.16	0.32	
			Swing low, T_A = –40°C to +85°C			0.45	
		V+ = 15 V, R_L = 600 Ω to V+ / 2	Swing high	13.4	14.1		
			Swing high, T_A = –40°C to +85°C	13			
			Swing low		0.5	1	
			Swing low, T_A = –40°C to +85°C			1.3	
I_{SC}	Output short-circuit current	V+ = 5 V, sourcing, V_O = 0 V		16	20		mA
			T_A = –40°C to +85°C	12			
		V+ = 5 V, sinking, V_O = 5 V		11	15		
			T_A = –40°C to +85°C	9.5			
		V+ = 15 V, sourcing, V_O = 0 V		28	30		
			T_A = –40°C to +85°C	22			
		V+ = 15 V, sinking, V_O = 12 V(1)		30	30		
			T_A = –40°C to +85°C	24			

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5.5 Electrical Characteristics: V_S = 5 V (continued)

at T_J = +25°C, V+ = 5 V, V– = 0 V, V_{CM} = V_{OUT} = V+ / 2, and R_L > 1 MΩ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
I_S	Supply current	Per amplifier, V+ = 5 V, V_O = V+ / 2			0.5	0.7	mA
			T_A = –40°C to +85°C			0.9	
		Per amplifier, V+ = 15 V, V_O = V+ / 2			0.65	0.8	
			T_A = –40°C to +85°C			0.95	
AC SPECS							
SR	Slew rate(2)	LMC6482AI V+ = 15 V, 10-V step		1	1.3		V/µs
			T_A = –40°C to +85°C	0.7			
		LMC6482I V+ = 15 V, 10-V step		0.9	1.3		
			T_A = –40°C to +85°C	0.63			
GBW	Gain bandwidth	V+ = 15 V			1.5		MHz
Θ_m	Phase margin				50		Deg
G_m	Gain margin				15		dB
	Amp-to-amp isolation	V+ = 15 V, R_L = 100 kΩ to 7.5 V, V_O = 12 V_{PP}, f = 1 kHz			150		dB
e_n	Input-referred voltage noise	f = 1 kHz, V_{CM} = 1 V			37		nV/√Hz
i_n	Input current noise density	f = 1 kHz			0.03		pA/√Hz
THD	Total harmonic distortion	f = 10 kHz, A_V = –2, R_L = 10 kΩ	V_O = 8.5 V_{PP}		0.01		%
			V+ = 10 V, V_O = 4.1 V_{PP}		0.01		

(1) Do not short circuit output to V+, when V+ is greater than 13 V or reliability is adversely affected.
(2) Number specified is the slower of either the positive or negative slew rates.

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5.6 Electrical Characteristics: V_S = 3 V

at T_J = +25°C, V+ = 3 V, V– = 0 V, V_{CM} = V_{OUT} = V+ / 2, and R_L > 1 MΩ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
DC SPECS							
V_{OS}	Input offset voltage	LMC6482AI			±0.9	±2	mV
			T_A = –40°C to +85°C			±2.7	
		LMC6482I			±0.9	±3	
			T_A = –40°C to +85°C			±3.7	
dV_{OS}/dT	Input offset voltage drift	T_A = –40°C to +85°C			±2		µV/°C
I_B	Input bias current				±0.02		pA
I_{OS}	Input offset current				±0.01		pA
CMRR	Common-mode rejection ratio	0 V < V_{CM} < 3 V	LMC6482AI	64	74		dB
			LMC6482I	60	74		
PSRR	Power-supply rejection ratio	3 V < V+ < 15 V, V– = 0 V	LMC6482AI	68	80		dB
			LMC6482I	60	80		
V_{CM}	Input common-mode voltage	For CMRR ≥ 50 dB	Low		(V–) – 0.25	0	V
			High	(V+)	(V+) + 0.25		
V_O	Voltage output swing	R_L = 2 kΩ to V+ / 2	Swing high		2.8		V
			Swing low		0.2		
		R_L = 600 Ω to V+ / 2	Swing high	2.5	2.7		
			Swing low		0.37	0.6	
I_S	Supply current	Per amplifier			0.4125	0.6	mA
			T_A = –40°C to +85°C			0.75	
AC SPECS							
SR	Slew rate(1)	Voltage follower with 2-V step input			0.9		V/µs
GBW	Gain bandwidth				1		MHz
THD	Total harmonic distortion	f = 10 kHz, A_V = –2, R_L = 10 kΩ, V_O = 2 V_{PP}			0.01		%

(1) Number specified is the slower of either the positive or negative slew rates.

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5.7 Typical Characteristics

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

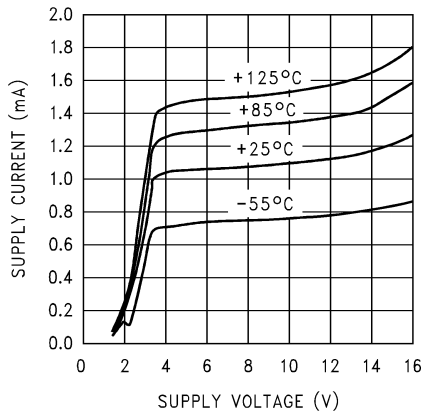


Figure 5-1. Supply Current vs Supply Voltage

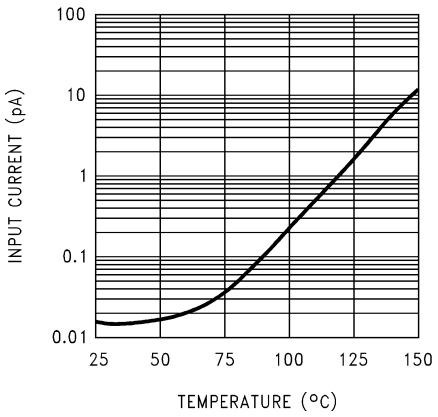


Figure 5-2. Input Current vs Temperature

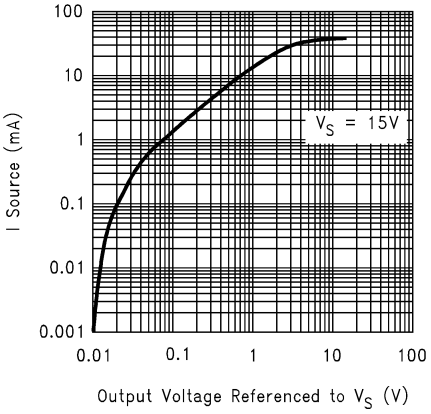


Figure 5-3. Sourcing Current vs Output Voltage

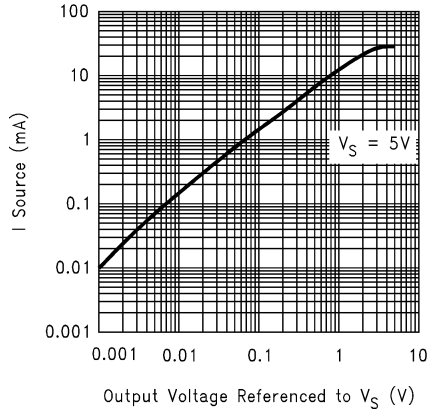


Figure 5-4. Sourcing Current vs Output Voltage

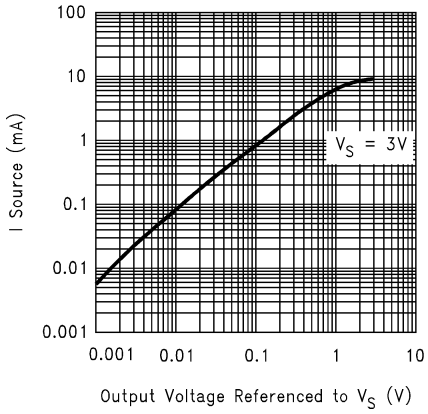


Figure 5-5. Sourcing Current vs Output Voltage

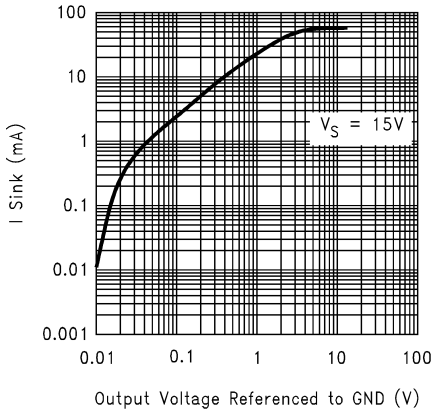


Figure 5-6. Sinking Current vs Output Voltage

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5.7 Typical Characteristics (continued)

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

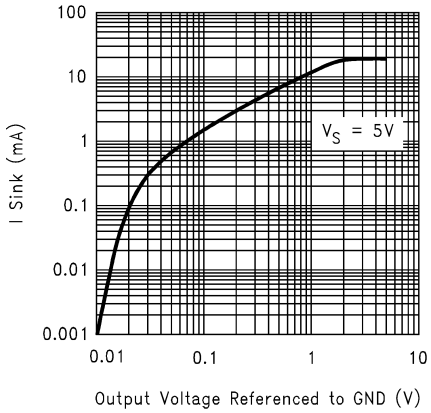


Figure 5-7. Sinking Current vs Output Voltage

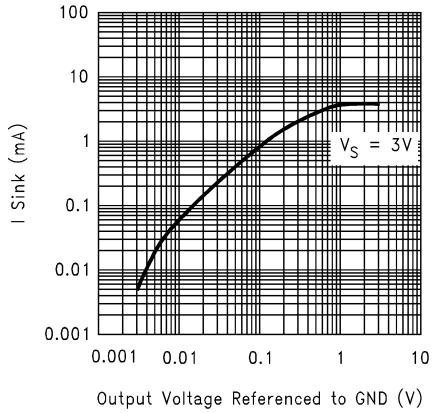


Figure 5-8. Sinking Current vs Output Voltage

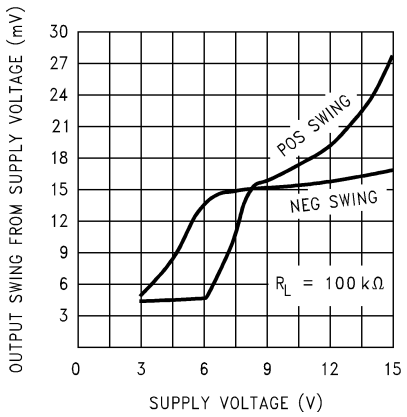


Figure 5-9. Output Voltage Swing vs Supply Voltage

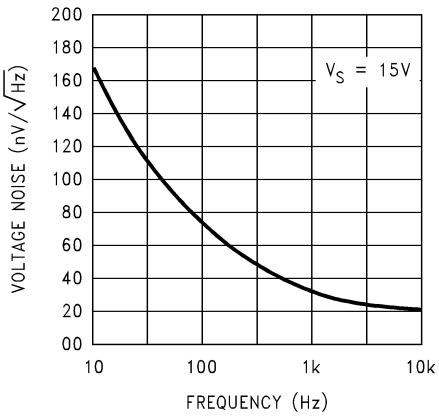


Figure 5-10. Input Voltage Noise vs Frequency

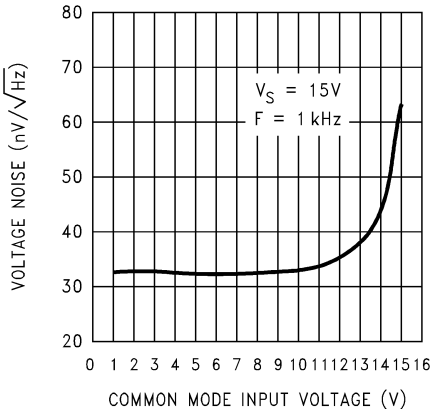


Figure 5-11. Input Voltage Noise vs Input Voltage

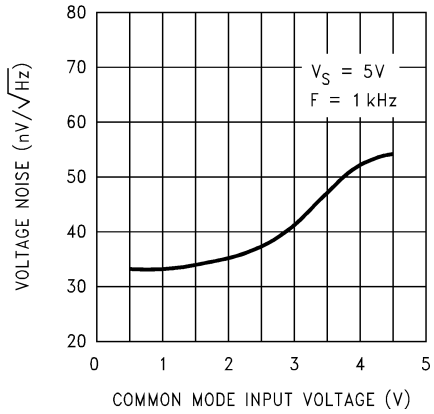


Figure 5-12. Input Voltage Noise vs Input Voltage

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5.7 Typical Characteristics (continued)

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

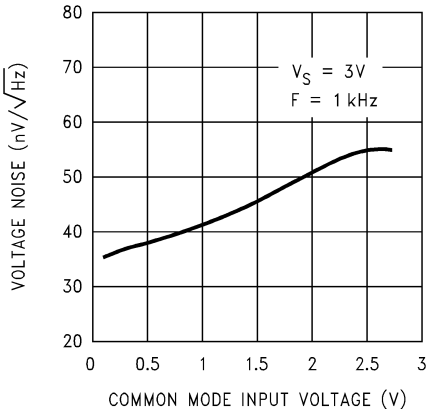


Figure 5-13. Input Voltage Noise vs Input Voltage

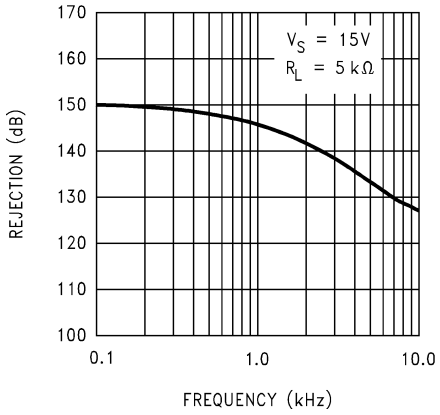


Figure 5-14. Crosstalk Rejection vs Frequency

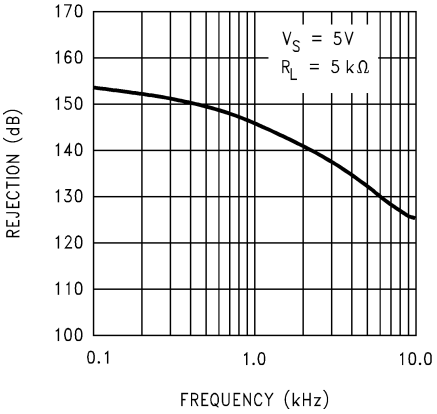


Figure 5-15. Crosstalk Rejection vs Frequency

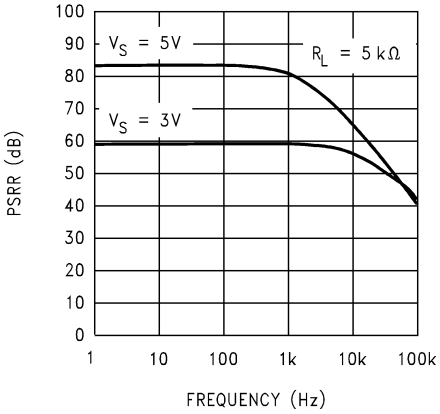


Figure 5-16. Positive PSRR vs Frequency

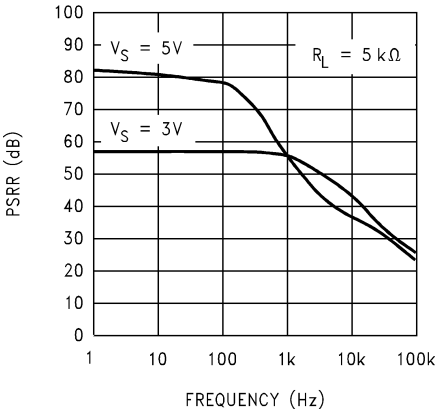


Figure 5-17. Negative PSRR vs Frequency

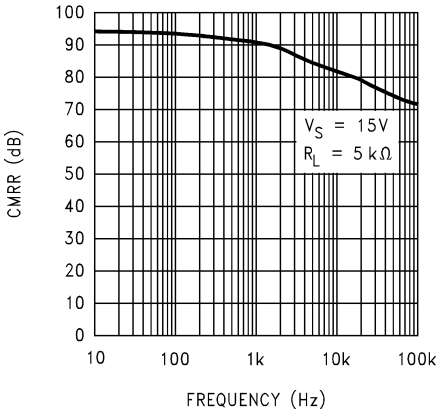


Figure 5-18. CMRR vs Frequency

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5.7 Typical Characteristics (continued)

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

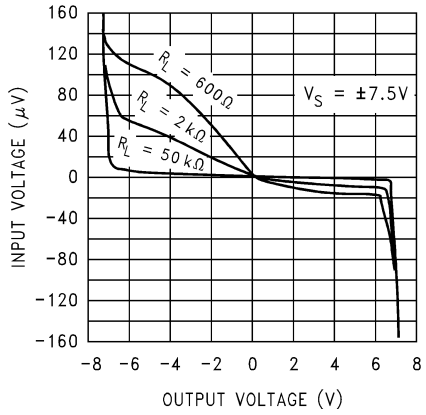


Figure 5-19. Input Voltage vs Output Voltage

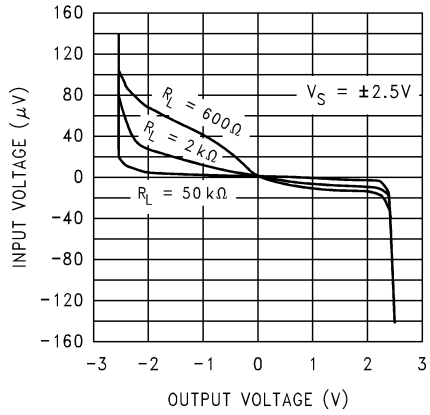


Figure 5-20. Input Voltage vs Output Voltage

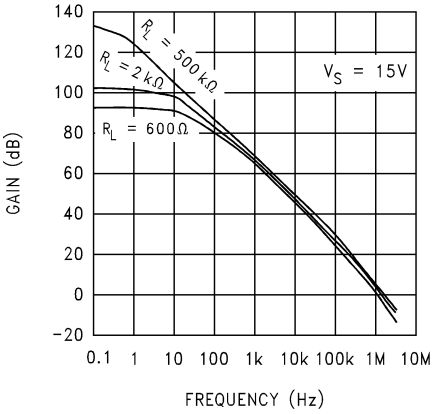


Figure 5-21. Open-Loop Frequency Response

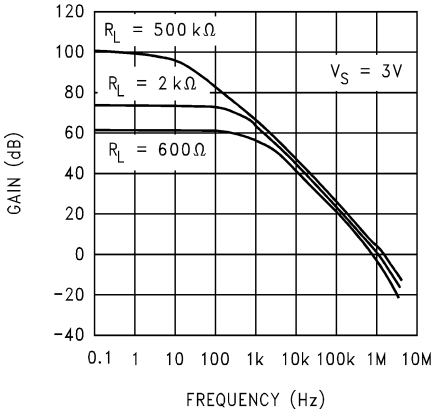


Figure 5-22. Open-Loop Frequency Response

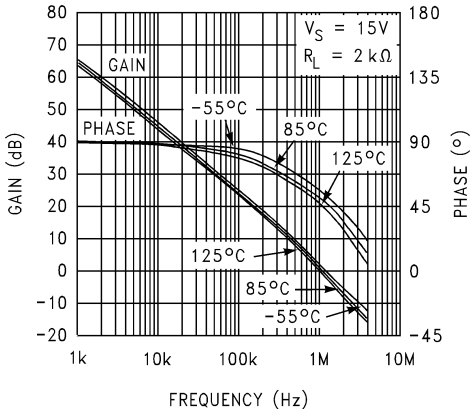


Figure 5-23. Open-Loop Frequency Response vs Temperature

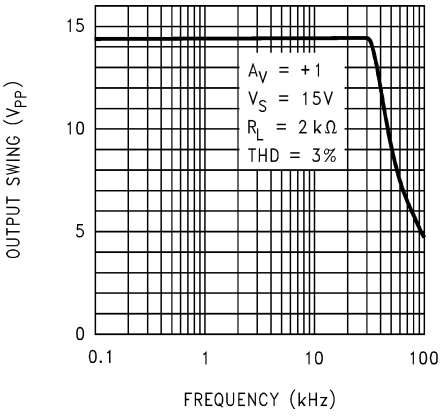


Figure 5-24. Maximum Output Swing vs Frequency

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5.7 Typical Characteristics (continued)

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

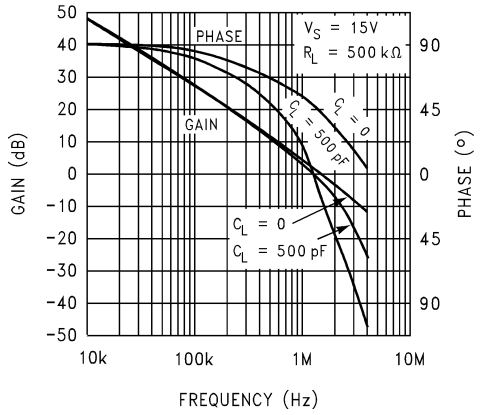


Figure 5-25. Gain and Phase vs Capacitive Load

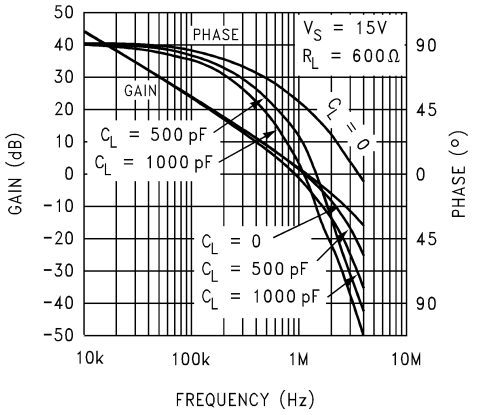


Figure 5-26. Gain and Phase vs Capacitive Load

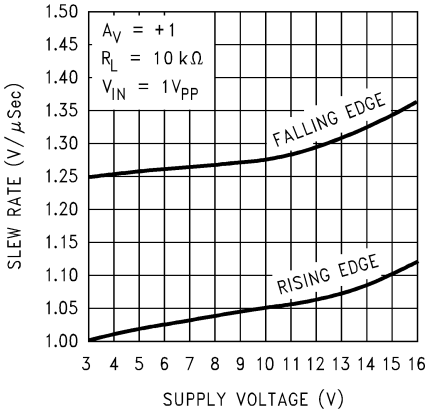


Figure 5-27. Slew Rate vs Supply Voltage

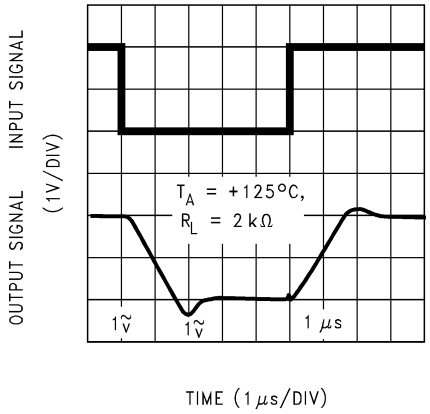


Figure 5-28. Noninverting Large Signal Pulse Response

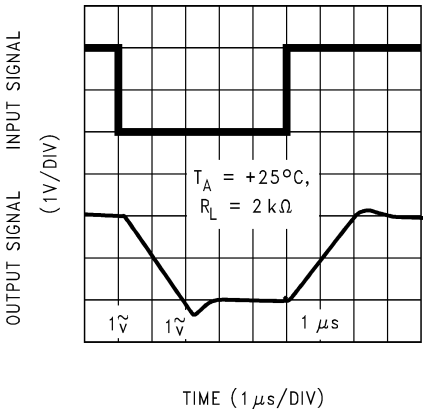


Figure 5-29. Noninverting Large Signal Pulse Response

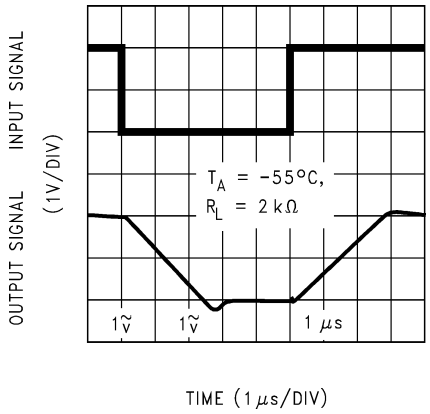


Figure 5-30. Noninverting Large Signal Pulse Response

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5.7 Typical Characteristics (continued)

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

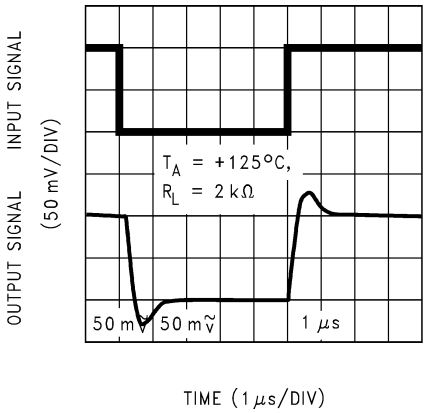


Figure 5-31. Noninverting Small Signal Pulse Response

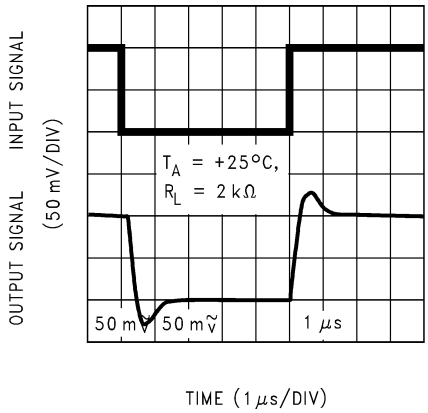


Figure 5-32. Noninverting Small Signal Pulse Response

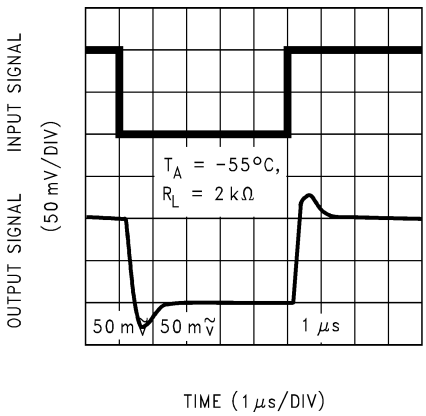


Figure 5-33. Noninverting Small Signal Pulse Response

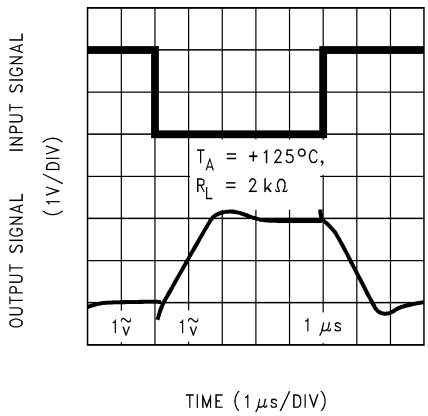


Figure 5-34. Inverting Large Signal Pulse Response

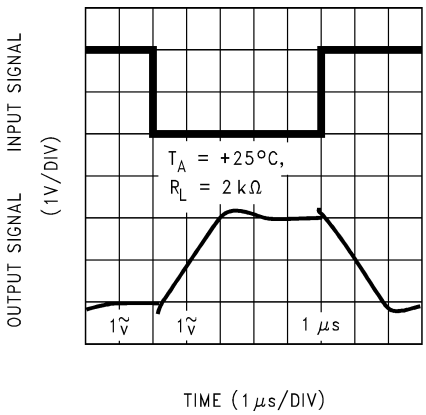


Figure 5-35. Inverting Large Signal Pulse Response

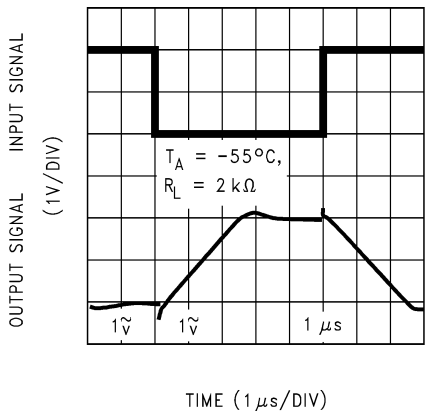


Figure 5-36. Inverting Large Signal Pulse Response

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5.7 Typical Characteristics (continued)

at V_S = 15 V, single supply, and T_A = 25°C (unless otherwise specified)

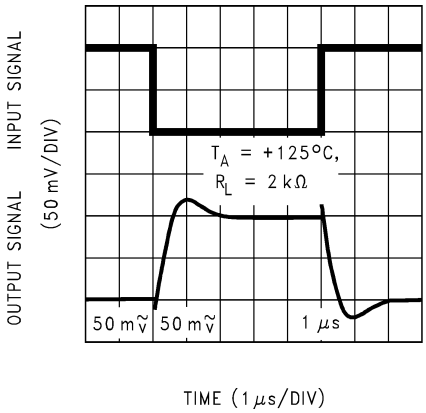


Figure 5-37. Inverting Small Signal Pulse Response

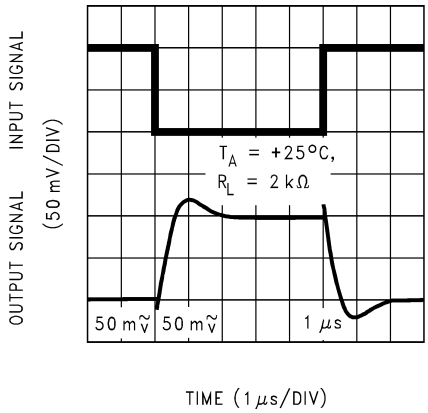


Figure 5-38. Inverting Small Signal Pulse Response

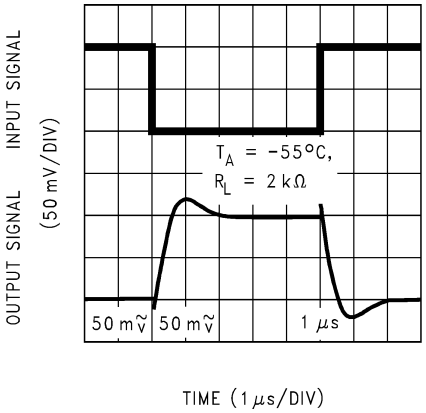


Figure 5-39. Inverting Small Signal Pulse Response

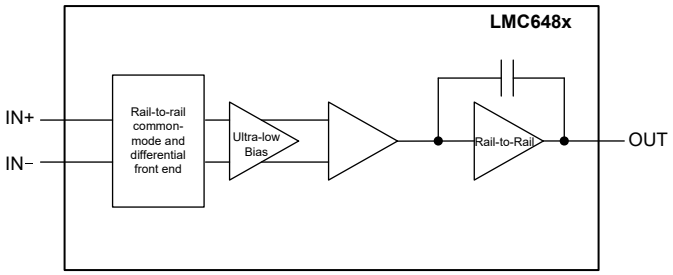
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6 Detailed Description

6.1 Overview

The LMC6482 is a CMOS operational amplifier that supports both rail-to-rail inputs and outputs. The device operates in both dual-supply mode and single-supply mode.

6.2 Functional Block Diagram



6.3 Feature Description

6.3.1 Amplifier Topology

The LMC6482 is a true rail-to-rail input operational amplifier with an input common-mode range that extends 300 mV beyond either supply rail. When the input common-mode voltage swings to about 3 V from the positive rail, some dc specifications, namely offset voltage, can be slightly degraded. Figure 6-1 illustrates this behavior. The LMC6482 incorporates a specially designed input stage to reduce the inherent accuracy problems seen in other rail-to-rail input amplifiers. The LMC6482 input stage design is complemented by an output stage capable of rail-to-rail output swing even when driving a large load. Rail-to-rail output swing is obtained by taking the output directly from the internal integrator instead of an output buffer stage.

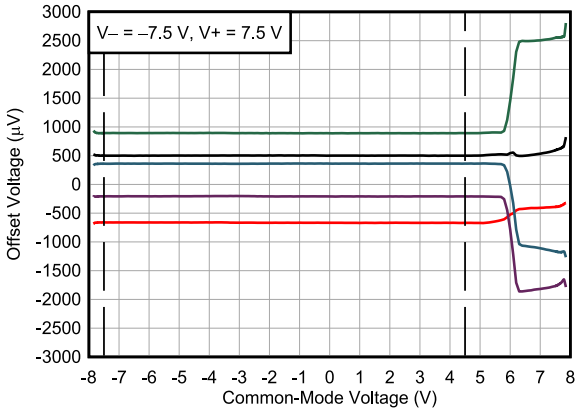


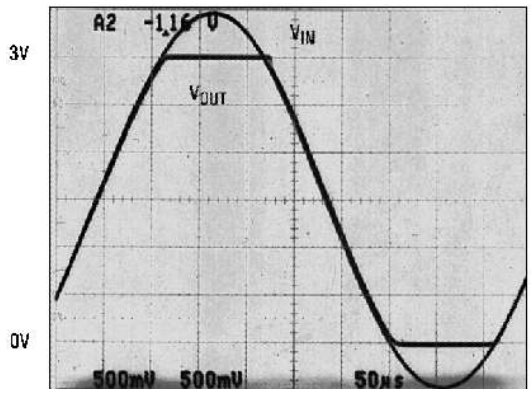
Figure 6-1. Input Offset Voltage vs Common-Mode Voltage

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6.3.2 Input Common-Mode Voltage Range

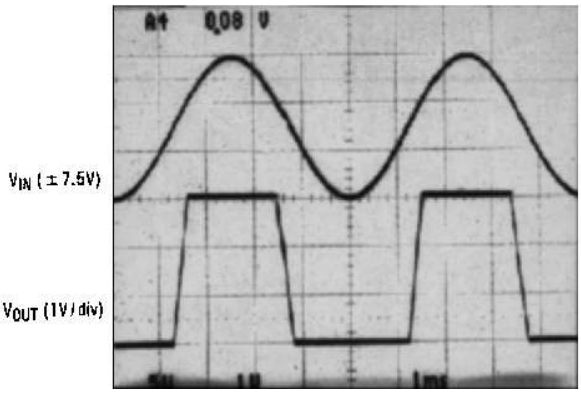
Unlike Bi-FET amplifier designs, the LMC6482 does not exhibit phase inversion when an input voltage exceeds the negative supply voltage. Figure 6-2 shows an input voltage exceeding both supplies with no resulting phase inversion on the output.

The absolute maximum input voltage is 300 mV beyond either supply rail at room temperature. Voltages greatly exceeding this absolute maximum rating, as in Figure 6-3, can cause excessive current to flow in or out of the input pins possibly affecting reliability.



An input voltage signal exceeds the LMC6482 power supply voltages with no output phase inversion.

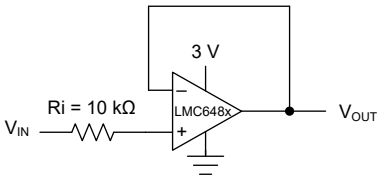
Figure 6-2. Input Voltage



A ±7.5-V input signal greatly exceeds the 3-V supply in Figure 6-4 causing no phase inversion due to R_I.

Figure 6-3. Input Signal

Applications that exceed this rating must externally limit the maximum input current to ±5 mA with an input resistor (R_I) as shown in Figure 6-4.



NOTE: R_I input current protection for voltages exceeding the supply voltages.

Figure 6-4. R_I Input Current Protection for Voltages Exceeding the Supply Voltages

6.3.3 Rail-to-Rail Output

The LMC6482 output can swing to within a few hundred millivolts of either supply voltage. Use the specified output swing specifications to calculate an approximate output resistance for different sourcing and sinking conditions. Use the calculated output resistance to estimate the maximum output voltage swing as a function of load.

6.4 Device Functional Modes

The LMC6482 can be used in applications where each amplifier channel is used independently, or in applications in which the channels are cascaded. See the *Section 7.2* section for more information.

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7 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

7.1 Application Information

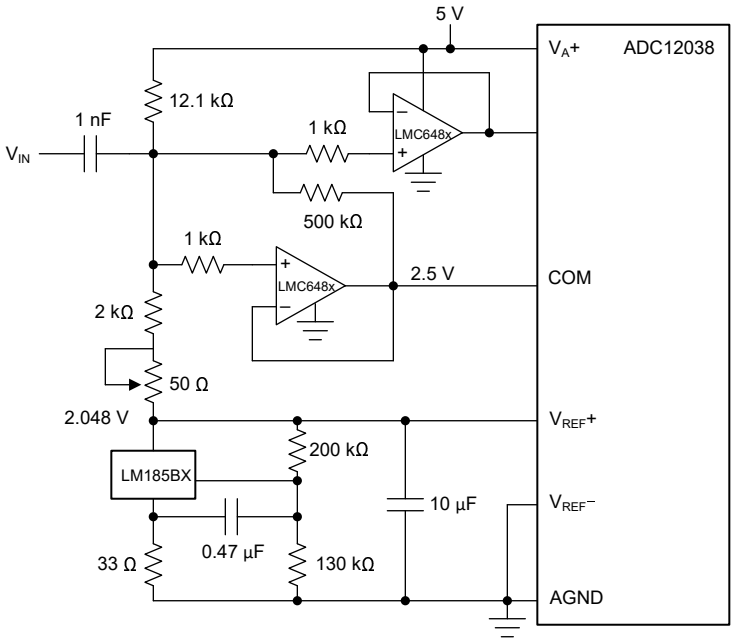
7.1.1 Upgrading Applications

The LMC6482 have industry-standard pin outs to retrofit existing applications. System performance can be greatly increased by the features of the LMC6482. The key benefit of designing in the LMC6482 is increased linear signal range. Most op-amps have limited input common-mode ranges. Signals that exceed this range generate a nonlinear output response that persists long after the input signal returns to the common-mode range.

Linear signal range is vital in applications such as filters where signal peaking can exceed input common-mode ranges resulting in output phase inversion or severe distortion.

7.1.2 Data Acquisition Systems

Figure 7-1 shows a low-power, single-supply data-acquisition system achieved by buffering the ADC12038 with the LMC6482. Capable of using the full supply range, the LMC6482 does not require input signals to be scaled down to meet limited common-mode voltage ranges. The LMC6482 CMRR of 82 dB maintains integral linearity of a 12-bit data acquisition system to ±0.325 LSB. Other rail-to-rail input amplifiers with only 50 dB of CMRR can degrade the accuracy of the data acquisition system to only 8 bits.



NOTE: Operating from the same supply voltage, the LMC6482 buffers the ADC12038 maintaining excellent accuracy.

Figure 7-1. Buffering the ADC12038 With the LMC648x

7.1.3 Instrumentation Circuits

The LMC6482 have high input impedance, large common-mode range and high CMRR needed for designing instrumentation circuits. Instrumentation circuits designed with the LMC6482 can reject a larger range of common-mode signals than most in-amps. This makes instrumentation circuits designed with the LMC6482 an excellent choice for noisy or industrial environments. Other applications that benefit from these features include analytic medical instruments, magnetic field detectors, gas detectors, and silicon-based transducers.

A small valued potentiometer is used in series with R_g to set the differential gain of the 3-op-amp instrumentation circuit in Figure 7-2. This combination is used instead of one large valued potentiometer to increase gain trim accuracy and reduce error due to vibration. An improved design that can help increase accuracy, save cost, and reduce board space can be achieved by using the RES11A matched resistor pair series.

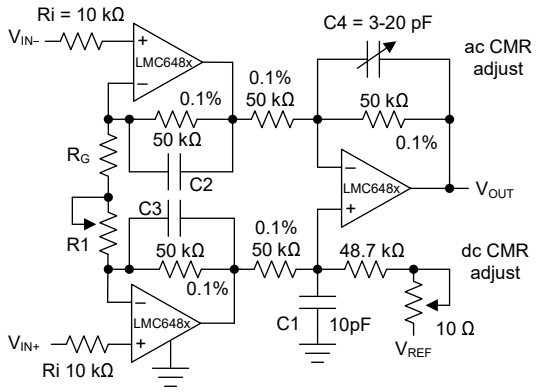


Figure 7-2. Low-Power, Three-Op-Amp Instrumentation Amplifier

The Figure 7-3 shows how a high precision, high CMRR, and low drift in-amp can be achieved using two matched resistor pairs. Using a 1:4 ratio, a gain of 36 V/V can be easily implemented. Other gain options are possible by using the various ratios available. One downside to the original implementation in Figure 7-2 is that very high performance, 0.01% resistors and a couple of potentiometers are needed to achieve very high common-mode rejection and gain accuracy. High accuracy resistors can be very expensive and add to board layout size and complexity. Another downside is that the temperature drift of the discrete resistors causes an increase in gain error that is not easily calibrated out.

The RES11A matched resistor pairs provide high common-mode rejection and gain-error performance due to excellent matching to less than 0.05%. The resistors are on the same substrate; therefore, the resistors drift in the same direction, thereby minimizing temperature-related errors such as gain error drift. For a more detailed analysis of the benefits of the RES11A over discrete resistors, see the *Optimizing CMRR in Differential Amplifier Circuits With Precision Matched Resistor Divider Pairs* application note.

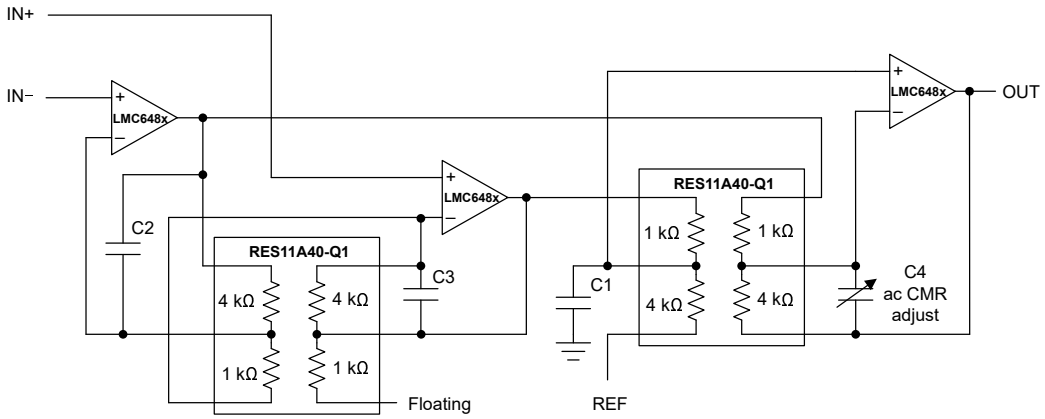


Figure 7-3. Improved Low-Power, Three-Op-Amp Instrumentation Amplifier With RES11A

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A two-op-amp instrumentation amplifier designed for a gain of 100 V/V is shown in Figure 7-4. Low sensitivity trimming is made for offset voltage, CMRR, and gain. Low cost and low power consumption are the main advantages of this two-op-amp circuit. An alternative circuit with a gain of 10 V/V with the RES11A is also provided for this circuit in Figure 7-5.

Higher frequency and larger common-mode range applications are best facilitated by a three-op-amp instrumentation amplifier.

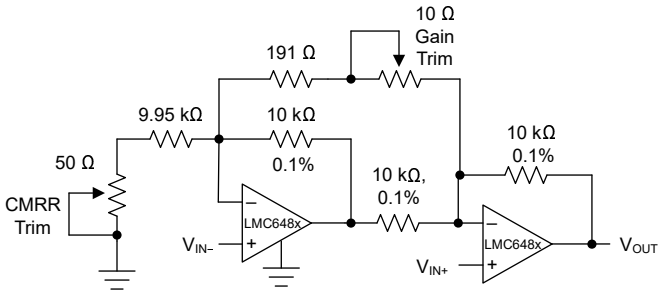


Figure 7-4. Low-Power, Two-Op-Amp Instrumentation Amplifier

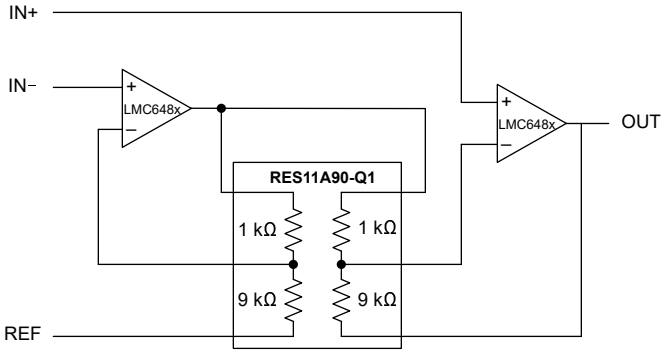


Figure 7-5. Low-Power, Two-Op-Amp Instrumentation Amplifier with RES11A

7.2 Typical Applications

7.2.1 3-V Single-Supply Buffer Circuit

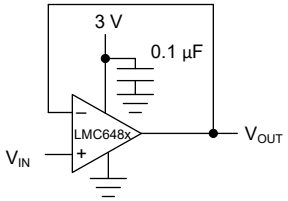


Figure 7-6. 3-V Single-Supply Buffer Circuit

7.2.1.1 Design Requirements

For best performance, ensure that the input voltage swing is between V+ and V–.

Also, ensure that the input does not exceed the common-mode input voltage range.

To reduce the risk of destabilizing the output, use resistive isolation on the output when driving capacitive loads (see *Section 7.2.1.2*).

When large feedback resistors are used, compensate for parasitic capacitance on the input, if necessary. See *Section 7.2.1.2*.

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7.2.1.2 Detailed Design Procedure

7.2.1.2.1 Capacitive Load Compensation

The LMC6482 provides a robust output stage for directly driving capacitive loads. Capacitive loads interact with the output impedance of the amplifier to create a pole that can cause instability. When driving capacitive loads, consider the closed-loop bandwidth and output impedance of the amplifier. The LMC6482 open-loop output impedance is shown in Figure 7-7.

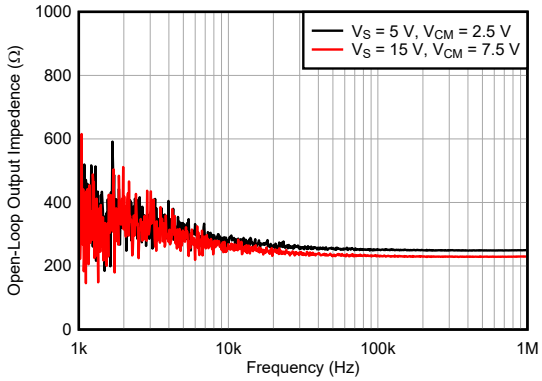


Figure 7-7. Open-Loop Output Impedance

In some applications, driving large capacitive loads is required and additional compensation is necessary. Capacitive load compensation can be accomplished using resistive isolation as shown in Figure 7-8. This simple technique is useful for isolating the capacitive inputs of multiplexers and A/D converters.

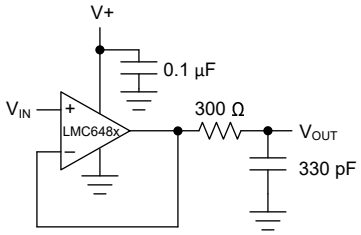


Figure 7-8. Resistive Isolation of a 330-pF Capacitive Load

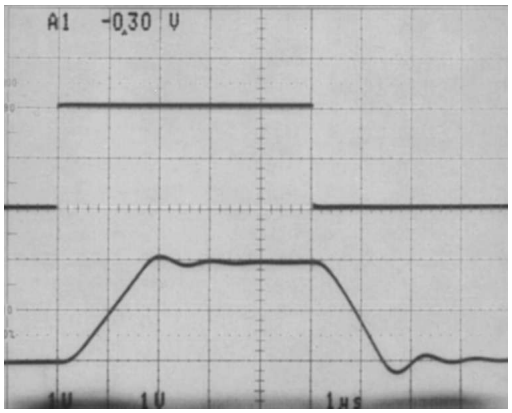


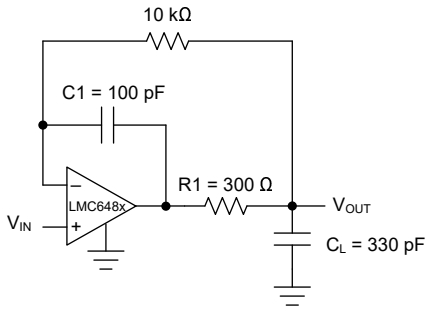
Figure 7-9. Pulse Response of the LMC6482 Circuit in Figure 7-8

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7.2.1.2.2 Capacitive Load Tolerance

The LMC6482 can typically directly drive a 100-pF load with V_S = 15 V at unity gain without oscillating. The unity gain follower is the most sensitive configuration. Direct capacitive loading reduces the phase margin of op-amps. The combination of the output impedance of the op-amp and the capacitive load induces phase lag. This results in either an underdamped pulse response or oscillation.

Figure 7-10 shows how improved frequency response is achieved by indirectly driving capacitive loads.



NOTE: Compensated to handle a 330-pF capacitive load.

Figure 7-10. LMC6482 Noninverting Amplifier

R1 and C1 serve to counteract the loss of phase margin by feeding forward the high-frequency component of the output signal back to the amplifiers inverting input, thereby preserving phase margin in the overall feedback loop. The values of R1 and C1 are experimentally determined for the desired pulse response. Figure 7-11 shows the resulting pulse response.

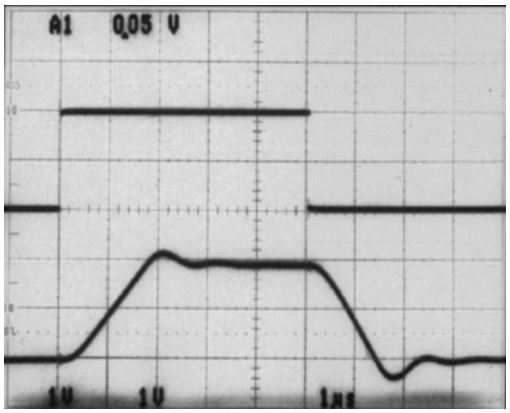


Figure 7-11. Pulse Response of LMC6482 Circuit in Figure 7-10

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7.2.1.2.3 Compensating For Input Capacitance

The use of large values of feedback resistance with amplifiers that have ultra-low input current, like the LMC6482, is quite common. Large feedback resistors can react with small values of input capacitance due to transducers, photo diodes, and circuits board parasitics to reduce phase margins.

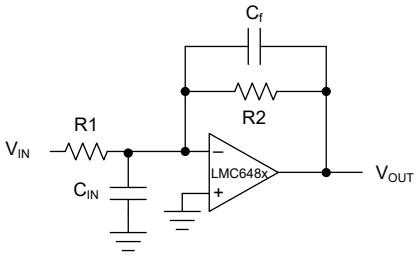


Figure 7-12. Canceling the Effect of Input Capacitance

The effect of input capacitance can be compensated for by adding a feedback capacitor. The feedback capacitor (as in Figure 7-12), C_f, is first estimated by:

$$\frac{1}{2\pi R1 C_{IN}} \geq \frac{1}{2\pi R2 C_f} \quad (1)$$

or

$$R1\ C_{IN} \leq R2\ C_f \quad (2)$$

which typically provides significant overcompensation.

Printed-circuit-board stray capacitance can be larger or smaller than that of a bread-board, so the actual optimum value for C_f can be different. Check the value of C_f on the actual circuit. (Refer to the LMC660 quad CMOS amplifier data sheet for a more detailed discussion.)

7.2.1.2.4 Offset Voltage Adjustment

Offset voltage adjustment circuits are illustrated in Figure 7-13 and Figure 7-14. Large value resistances and potentiometers are used to reduce power consumption while providing typically ±2.5 mV of adjustment range, referred to the input, for both configurations with V_S = ±5 V.

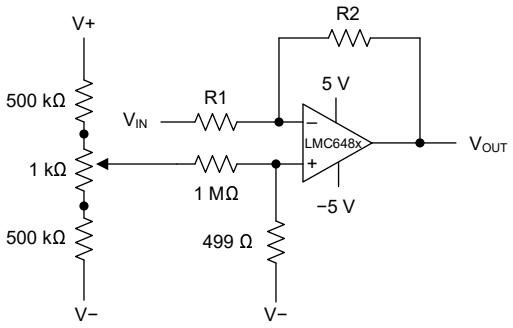


Figure 7-13. Inverting Configuration Offset Voltage Adjustment

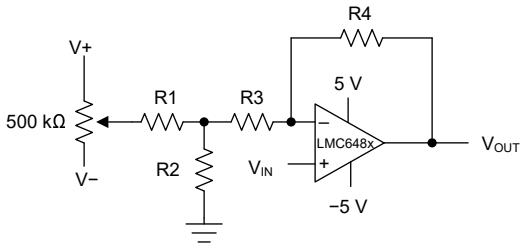


Figure 7-14. Noninverting Configuration Offset Voltage Adjustment

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7.2.1.3 Application Curves

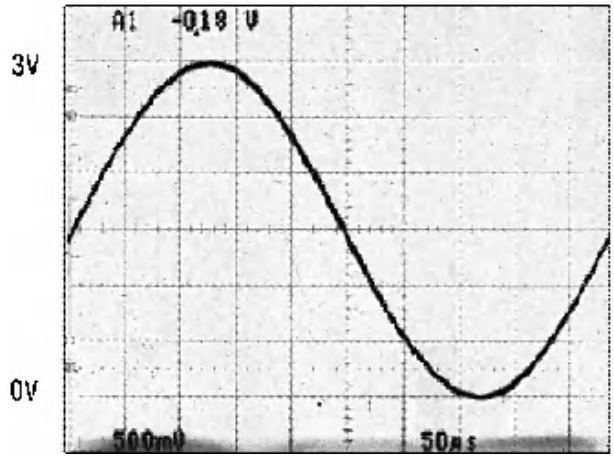


Figure 7-15. Rail-to-Rail Input

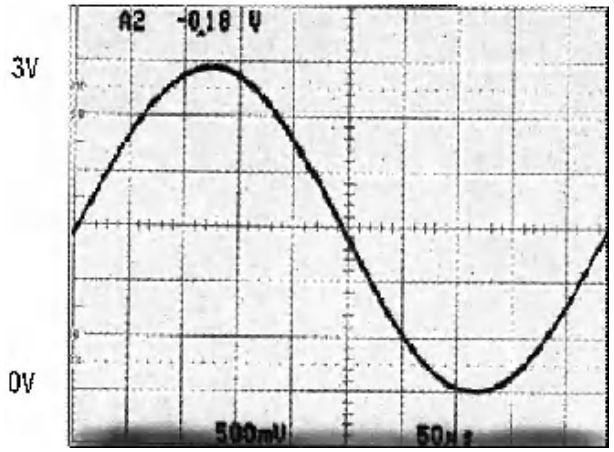


Figure 7-16. Rail-to-Rail Output

7.2.2 Typical Single-Supply Applications

The circuit in Figure 7-17 uses a single supply to half-wave rectify a sinusoid centered about ground. R_I limits current into the amplifier caused by the input voltage exceeding the supply voltage. Full-wave rectification is provided by the circuit in Figure 7-19.

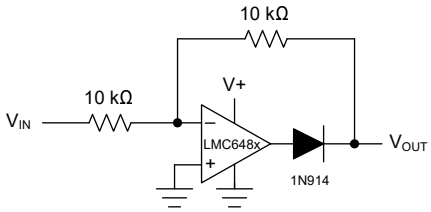


Figure 7-17. Half-Wave Rectifier With Input Current Protection (R_I)

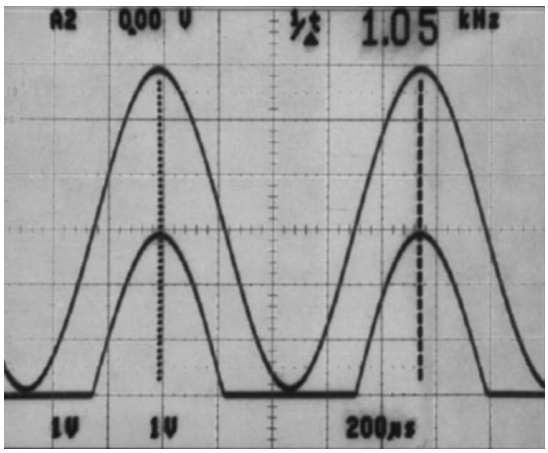


Figure 7-18. Half-Wave Rectifier Waveform

In Figure 7-23, dielectric absorption and leakage is minimized by using a polystyrene or polyethylene hold capacitor. The droop rate is primarily determined by the value of C_H and diode leakage current. The ultra-low input current of the LMC6482 has a negligible effect on droop. For applications requiring ultra-low input bias current, see the OPA928.

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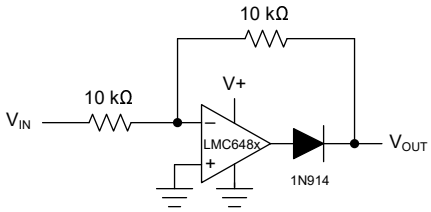


Figure 7-19. Full-Wave Rectifier With Input Current Protection (R_I)

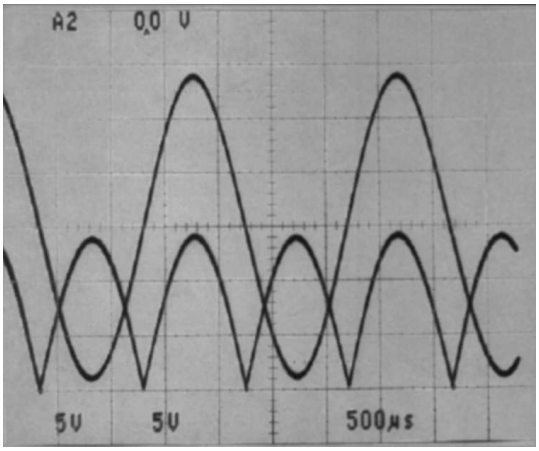


Figure 7-20. Full-Wave Rectifier Waveform

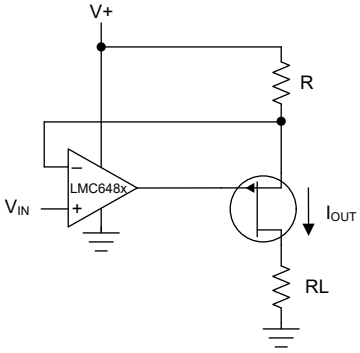


Figure 7-21. Large Compliance Range Current Source

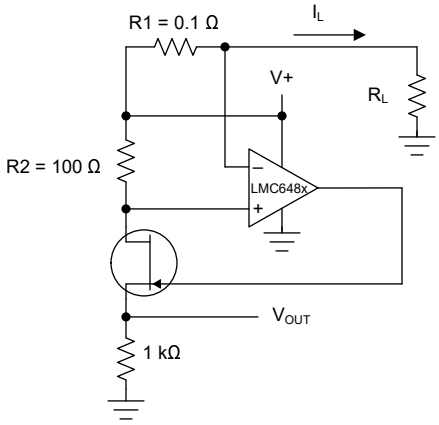


Figure 7-22. Positive Supply Current Sense

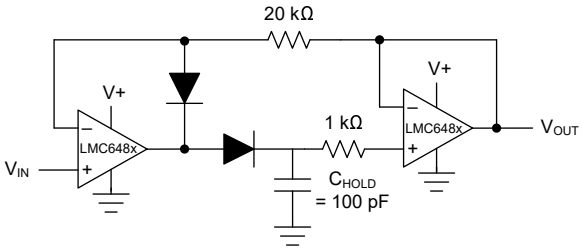


Figure 7-23. Low-Voltage Peak Detector With Rail-To-Rail Peak Capture Range

The high CMRR (82 dB) of the LMC6482 allows excellent accuracy throughout the rail-to-rail dynamic capture range of the circuit.

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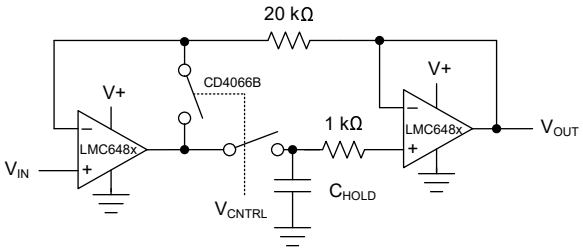


Figure 7-24. Rail-to-Rail Sample and Hold

The low-pass filter circuit in Figure 7-25 can be used as an anti-aliasing filter with the same voltage supply as the A/D converter.

Filter designs can also take advantage of the LMC6482 ultra-low input current. The ultra-low input current yields negligible offset error even when large value resistors are used. This in turn allows the use of smaller valued capacitors that take less board space and cost less.

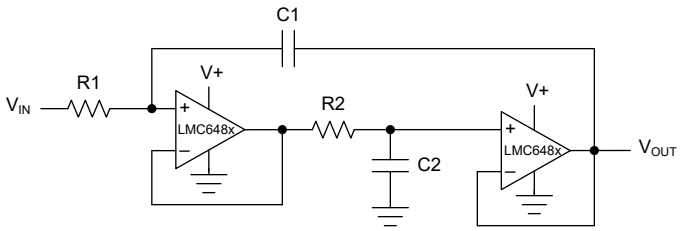


Figure 7-25. Rail-to-Rail, Single-Supply Low-Pass Filter

$$R1 = R2,\ C1 = C2,\ f = \frac{1}{2\pi R1C1},\ DF = \frac{1}{2}\sqrt{\frac{C2}{C1}}\sqrt{\frac{R2}{R1}} \quad (3)$$

7.3 Power Supply Recommendations

The LMC6482 operates over a supply range of 3 V to 15.5 V. To achieve noise immunity as appropriate to the application, use good printed circuit board (PCB) layout practices for power-supply rails and planes, as well as bypass capacitors connected between the power-supply pins and ground.

7.4 Layout

7.4.1 Layout Guidelines

As a general rule, any circuit that must operate with less than 1000 pA of leakage current requires special layout of the PC board. To take advantage of the ultra-low input current of the LMC6482, typically less than 20 fA, an excellent layout is essential. Fortunately, the techniques of obtaining low leakages are quite simple. First, do not ignore the surface leakage of the PCB even though the leakage current can sometimes appear acceptably low, because under conditions of high humidity or dust or contamination, the surface leakage can be appreciable.

To minimize the effect of any surface leakage, lay out a ring of foil completely surrounding the LMC6482 inputs and the terminals of capacitors, diodes, conductors, resistors, relay terminals, and so forth connected to the inputs of the op amp, as in Figure 7-26. To have a significant effect, place guard rings on both the top and bottom of the PCB. This PC foil must then be connected to a voltage that is at the same voltage as the amplifier inputs, because no leakage current can flow between two points at the same potential. For example, a PCB trace-to-pad resistance of 10^{12} Ω, which is normally considered a very large resistance, can leak 5 pA if the trace were a 5-V bus adjacent to the pad of the input. This leakage can cause a 250 times degradation from the actual performance of the LMC6482. However, if a guard ring is held within 5 mV of the inputs, then even a resistance of 10^{11} Ω causes only 0.05 pA of leakage current. See Figure 7-27 through Figure 7-29 for typical connections of guard rings for standard op-amp configurations.

Be aware that when laying out a PCB for the sake of just a few circuits is not practical, another technique is even better than a guard ring on a PCB: Do not insert the input pin of the amplifier into the PCB at all, but bend the pin up in the air, and use only air as an insulator. Air is an excellent insulator. In this case you forgo some of the advantages of PCB construction, but the advantages are sometimes well worth the effort of using point-to-point up-in-the-air wiring. See Figure 7-30.

7.4.2 Layout Example

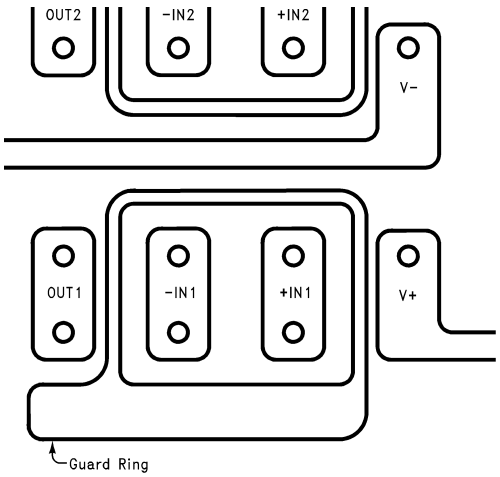


Figure 7-26. Example of Guard Ring in PCB Layout Typical Connections of Guard Rings

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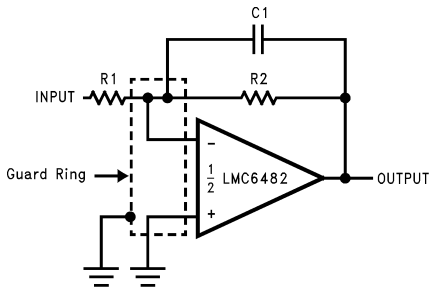


Figure 7-27. Inverting Amplifier Typical Connections of Guard Rings

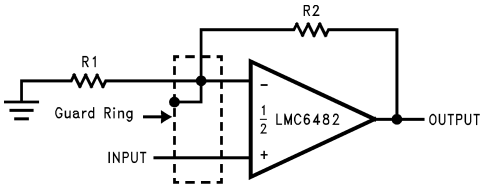


Figure 7-28. Noninverting Amplifier Typical Connections of Guard Rings

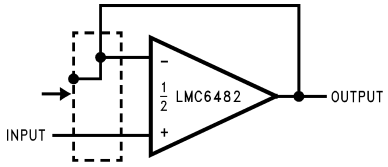
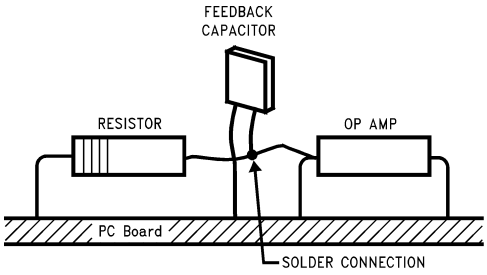


Figure 7-29. Follower Typical Connections of Guard Rings



Input pins are lifted out of PCB and soldered directly to components. All other pins connected to PCB.

Figure 7-30. Air Wiring

8 Device and Documentation Support

8.1 Device Support

8.1.1 Development Support

8.1.1.1 Spice Macromodel

A spice macromodel is available for the LMC6482. This model includes accurate simulation of the following:

- Input common-mode voltage range
- Frequency and transient response
- Gain bandwidth (GBW) dependence on loading conditions
- Quiescent and dynamic supply current
- Output swing dependence on loading conditions

8.1.1.2 PSpice® for TI

PSpice® for TI is a design and simulation environment that helps evaluate performance of analog circuits. Create subsystem designs and prototype solutions before committing to layout and fabrication, reducing development cost and time to market.

8.1.1.3 TINA-TI™ Simulation Software (Free Download)

TINA-TI™ simulation software is a simple, powerful, and easy-to-use circuit simulation program based on a SPICE engine. TINA-TI simulation software is a free, fully-functional version of the TINA™ software, preloaded with a library of macromodels, in addition to a range of both passive and active models. TINA-TI simulation software provides all the conventional dc, transient, and frequency domain analysis of SPICE, as well as additional design capabilities.

Available as a free download from the Design tools and simulation web page, TINA-TI simulation software offers extensive post-processing capability that allows users to format results in a variety of ways. Virtual instruments offer the ability to select input waveforms and probe circuit nodes, voltages, and waveforms, creating a dynamic quick-start tool.

Note

These files require that either the TINA software or TINA-TI software be installed. Download the free TINA-TI simulation software from the TINA-TI™ software folder.

8.1.1.4 DIP-Adapter-EVM

Speed up your op amp prototyping and testing with the DIP-Adapter-EVM, which provides a fast, easy and inexpensive way to interface with small, surface-mount devices. Connect any supported op amp using the included Samtec terminal strips or wire them directly to existing circuits. The DIP-Adapter-EVM kit supports the following industry-standard packages: D or U (SOIC-8), PW (TSSOP-8), DGK (VSSOP-8), DBV (SOT-23-6, SOT-23-5 and SOT-23-3), DCK (SC70-6 and SC70-5), and DRL (SOT563-6).

8.1.1.5 DIYAMP-EVM

The DIYAMP-EVM is a unique evaluation module (EVM) that provides real-world amplifier circuits, enabling the user to quickly evaluate design concepts and verify simulations. This EVM is available in three industry-standard packages (SC70, SOT23, and SOIC) and 12 popular amplifier configurations, including amplifiers, filters, stability compensation, and comparator configurations for both single and dual supplies.

8.1.1.6 TI Reference Designs

TI reference designs are analog solutions created by TI's precision analog applications experts. TI reference designs offer the theory of operation, component selection, simulation, complete PCB schematic and layout, bill of materials, and measured performance of many useful circuits. TI reference designs are available online at https://www.ti.com/reference-designs.

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8.1.1.7 Filter Design Tool

The filter design tool is a simple, powerful, and easy-to-use active filter design program. The filter design tool allows the user to create optimized filter designs using a selection of TI operational amplifiers and passive components from TI's vendor partners.

Available as a web-based tool from the Design tools and simulation web page, the filter design tool allows the user to design, optimize, and simulate complete multistage active filter solutions within minutes.

8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

8.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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PSpice® is a registered trademark of Cadence Design Systems, Inc.
All trademarks are the property of their respective owners.

8.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

8.6 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

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9 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision G (April 2020) to Revision H (November 2023)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document	1
• Deleted specifications are typical, high voltage gain, and power good output from *Features*	1
• Deleted M version device from data sheet; see the LMC6482QML for more information	1
• Updated front page figures in *Description*	1
• Updated *Pin Configuration and Functions*	2
• Added ± to input offset voltage, input offset voltage drift, input bias current, and input offset current in *Electrical Characteristics*	4
• Updated parameter names throughout *Electrical Characteristics* for consistency	4
• Deleted notes 1, 2, and 3 from *Electrical Characteristics*	4
• Changed supply current specification from total to per amplifier in *Electrical Characteristics*	4
• Deleted Figure 11 to 13, Figure 19 to 23, Figure 32 to 33, and Figure 47 to 52	8
• Updated functional block diagram	15
• Updated description of the input stage in *Amplifier Topology*	15
• Added Input Offset Voltage vs Common-Mode Voltage plot in *Amplifier Topology*	15
• Updated the description in *Rail-to-Rail Output*	16
• Added an improved instrumentation amplifier circuit to *Instrumentation Circuits*	18
• Added Figure 7-7, *Open-Loop Output Impedance* and related content to *Capacitive Load Compensation*	20
• Added OPA928 femtoampere-input bias-current op-amp recommendation to *Typical Single-Supply Applications*	23
• Deleted references to the library disk in *Spice Macromodel*	28

Changes from Revision F (April 2020) to Revision G (April 2020)	Page
• Deleted old note 4 from *Electrical Characteristics* for V+ = 5 V table	4

Changes from Revision E (April 2015) to Revision F (April 2020)	Page
• Changed junction temperature max value from –85°C to 85°C (typo) in *Recommended Operating Conditions* table	3

Changes from Revision D (March 2013) to Revision E (April 2015)	Page
• Added *Pin Configuration and Functions* section, *ESD Ratings* table, *Feature Description* section, *Device Functional Modes*, *Application and Implementation* section, *Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section	1

Changes from Revision C (March 2013) to Revision D (March 2013)	Page
• Changed layout of National Semiconductor Data Sheet to TI format	23

10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
LMC6482AIM	LIFEBUY	SOIC	D	8	95	Non-RoHS & Green	Call TI	Level-1-235C-UNLIM	-40 to 85	LMC64 82AIM	
LMC6482AIM/NOPB	LIFEBUY	SOIC	D	8	95	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 85	LMC64 82AIM	
LMC6482AIMX	LIFEBUY	SOIC	D	8	2500	Non-RoHS & Green	Call TI	Level-1-235C-UNLIM	-40 to 85	LMC64 82AIM	
LMC6482AIMX/NOPB	ACTIVE	SOIC	D	8	2500	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 85	LMC64 82AIM	Samples
LMC6482AIN/NOPB	ACTIVE	PDIP	P	8	40	RoHS & Green	NIPDAU	Level-1-NA-UNLIM	-40 to 85	LMC64 82AIN	Samples
LMC6482IM	LIFEBUY	SOIC	D	8	95	Non-RoHS & Green	Call TI	Level-1-235C-UNLIM	-40 to 85	LMC64 82IM	
LMC6482IM/NOPB	LIFEBUY	SOIC	D	8	95	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 85	LMC64 82IM	
LMC6482IMM	LIFEBUY	VSSOP	DGK	8	1000	Non-RoHS & Green	Call TI	Level-1-260C-UNLIM	-40 to 85	A10	
LMC6482IMM/NOPB	LIFEBUY	VSSOP	DGK	8	1000	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 85	A10	
LMC6482IMMX	LIFEBUY	VSSOP	DGK	8	3500	Non-RoHS & Green	Call TI	Level-1-260C-UNLIM	-40 to 85	A10	
LMC6482IMMX/NOPB	ACTIVE	VSSOP	DGK	8	3500	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 85	A10	Samples
LMC6482IMX	LIFEBUY	SOIC	D	8	2500	Non-RoHS & Green	Call TI	Level-1-235C-UNLIM	-40 to 85	LMC64 82IM	
LMC6482IMX/NOPB	ACTIVE	SOIC	D	8	2500	RoHS & Green	SN	Level-1-260C-UNLIM	-40 to 85	LMC64 82IM	Samples
LMC6482IN/NOPB	ACTIVE	PDIP	P	8	40	RoHS & Green	NIPDAU	Level-1-NA-UNLIM	-40 to 85	LMC6482IN	Samples

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".
RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.
Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

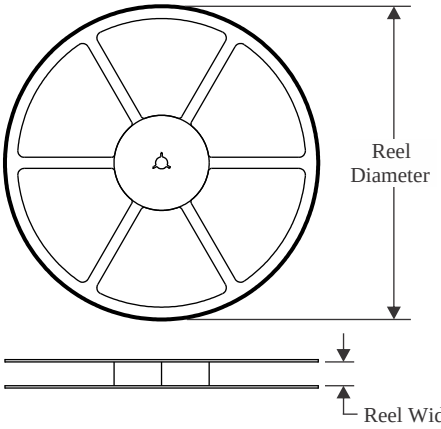
(6) Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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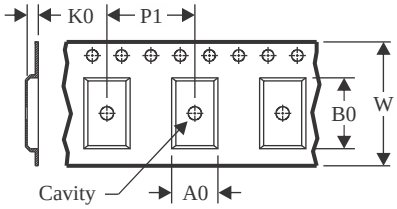
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

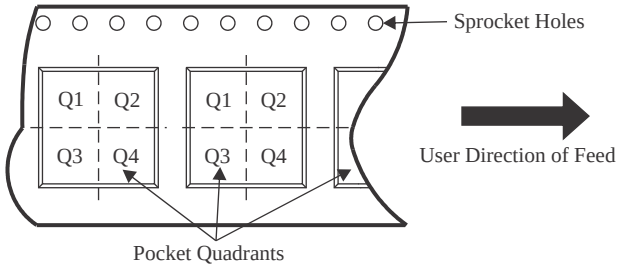


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

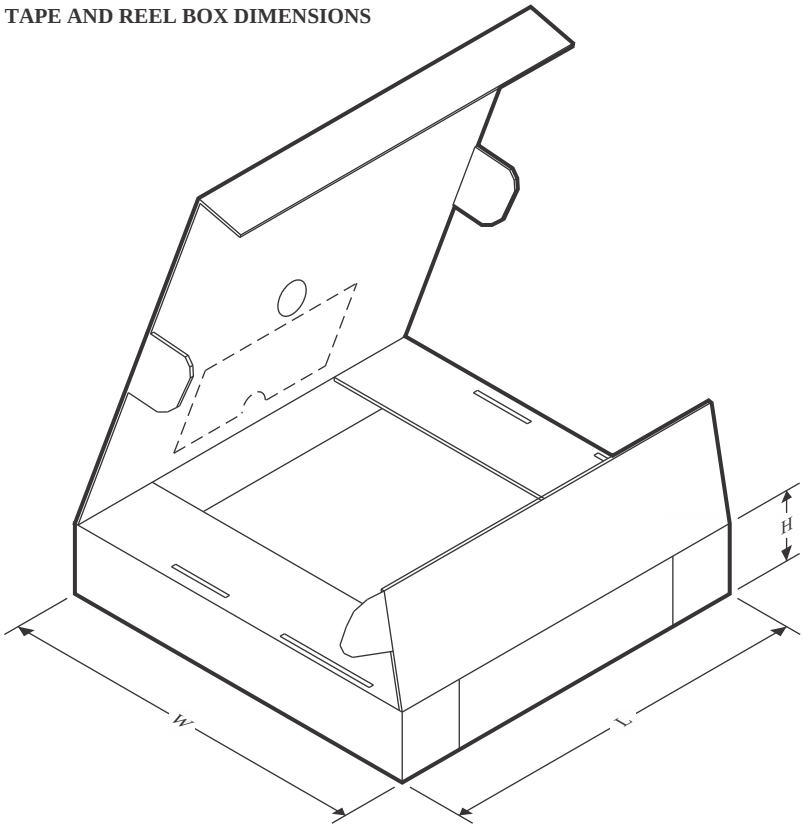
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
LMC6482AIMX	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1
LMC6482AIMX/NOPB	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1
LMC6482IMM	VSSOP	DGK	8	1000	178.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
LMC6482IMM/NOPB	VSSOP	DGK	8	1000	178.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
LMC6482IMMX	VSSOP	DGK	8	3500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
LMC6482IMMX/NOPB	VSSOP	DGK	8	3500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
LMC6482IMX	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1
LMC6482IMX/NOPB	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1

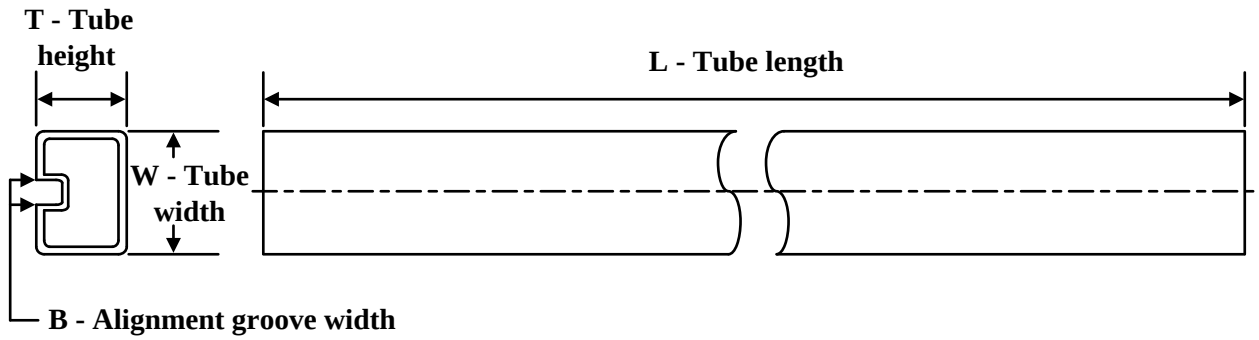
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

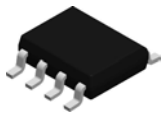
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
LMC6482AIMX	SOIC	D	8	2500	367.0	367.0	35.0
LMC6482AIMX/NOPB	SOIC	D	8	2500	367.0	367.0	35.0
LMC6482IMM	VSSOP	DGK	8	1000	208.0	191.0	35.0
LMC6482IMM/NOPB	VSSOP	DGK	8	1000	208.0	191.0	35.0
LMC6482IMMX	VSSOP	DGK	8	3500	367.0	367.0	35.0
LMC6482IMMX/NOPB	VSSOP	DGK	8	3500	367.0	367.0	35.0
LMC6482IMX	SOIC	D	8	2500	367.0	367.0	35.0
LMC6482IMX/NOPB	SOIC	D	8	2500	367.0	367.0	35.0

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
LMC6482AIM	D	SOIC	8	95	495	8	4064	3.05
LMC6482AIM	D	SOIC	8	95	495	8	4064	3.05
LMC6482AIM/NOPB	D	SOIC	8	95	495	8	4064	3.05
LMC6482AIN/NOPB	P	PDIP	8	40	502	14	11938	4.32
LMC6482IM	D	SOIC	8	95	495	8	4064	3.05
LMC6482IM	D	SOIC	8	95	495	8	4064	3.05
LMC6482IM/NOPB	D	SOIC	8	95	495	8	4064	3.05
LMC6482IN/NOPB	P	PDIP	8	40	502	14	11938	4.32

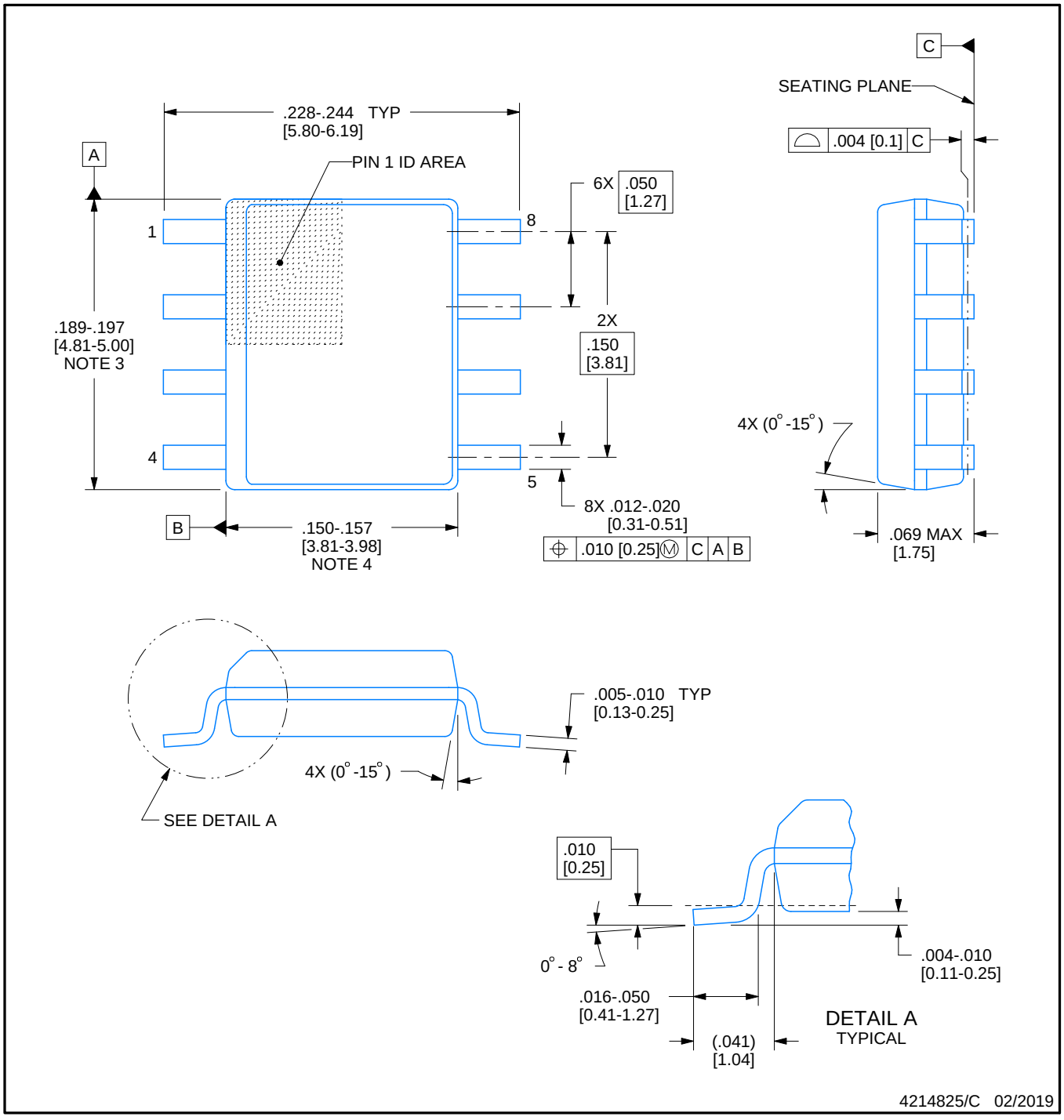


PACKAGE OUTLINE

D0008A

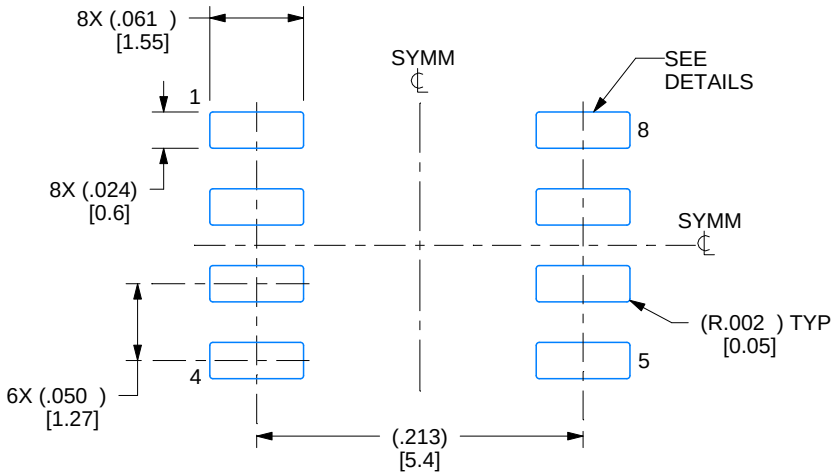
SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

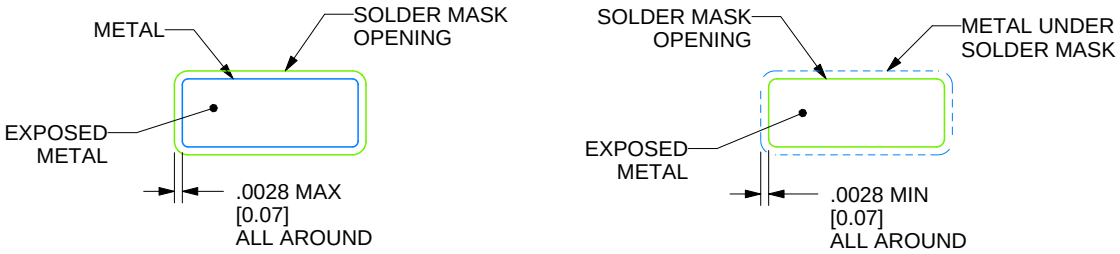


NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

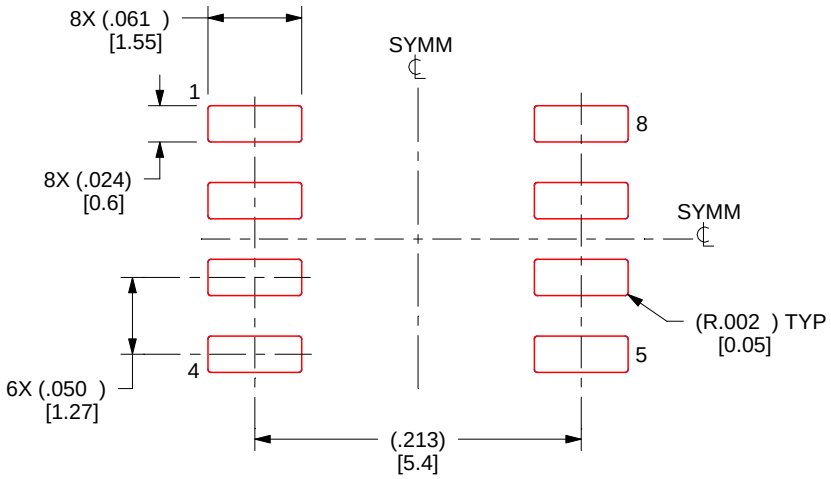
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



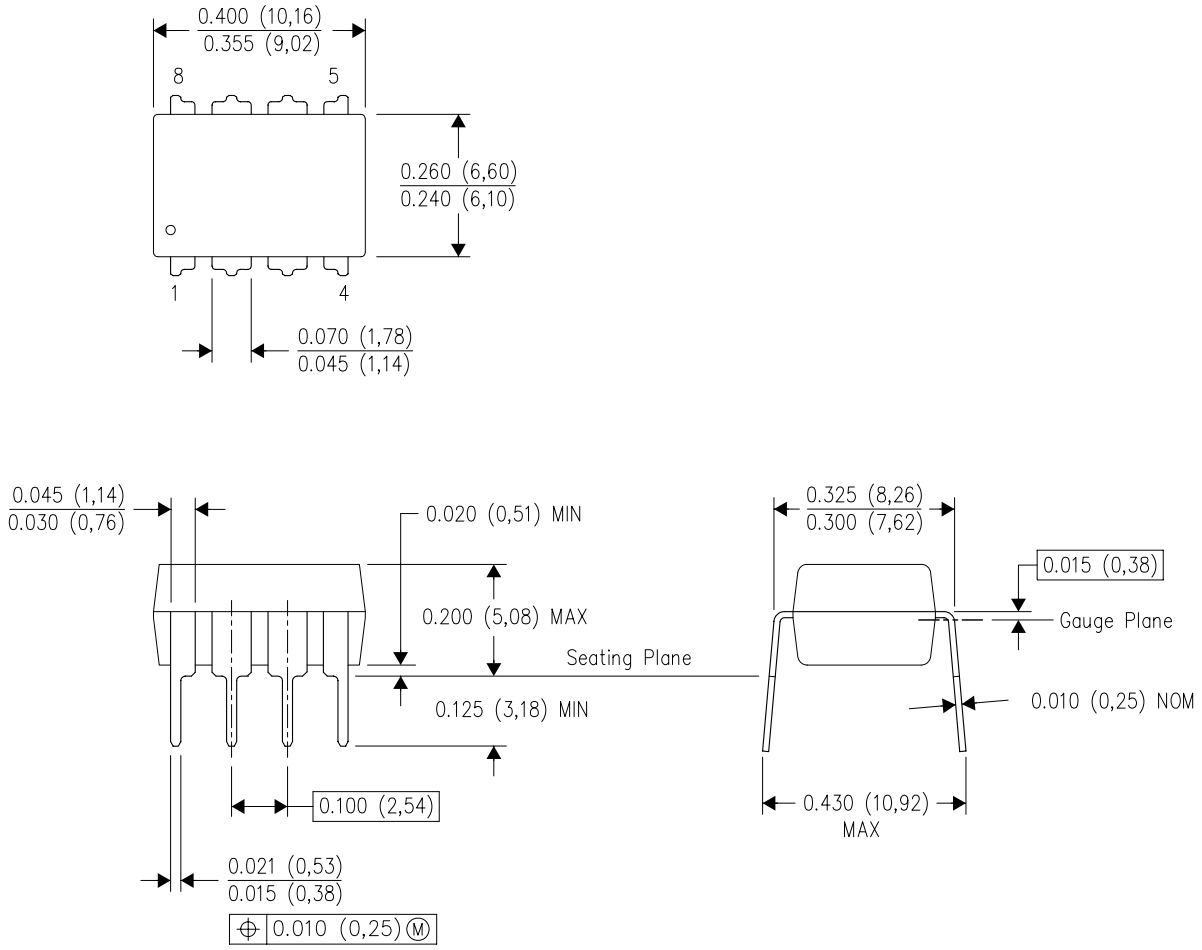
SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

4214825/C 02/2019

NOTES: (continued)

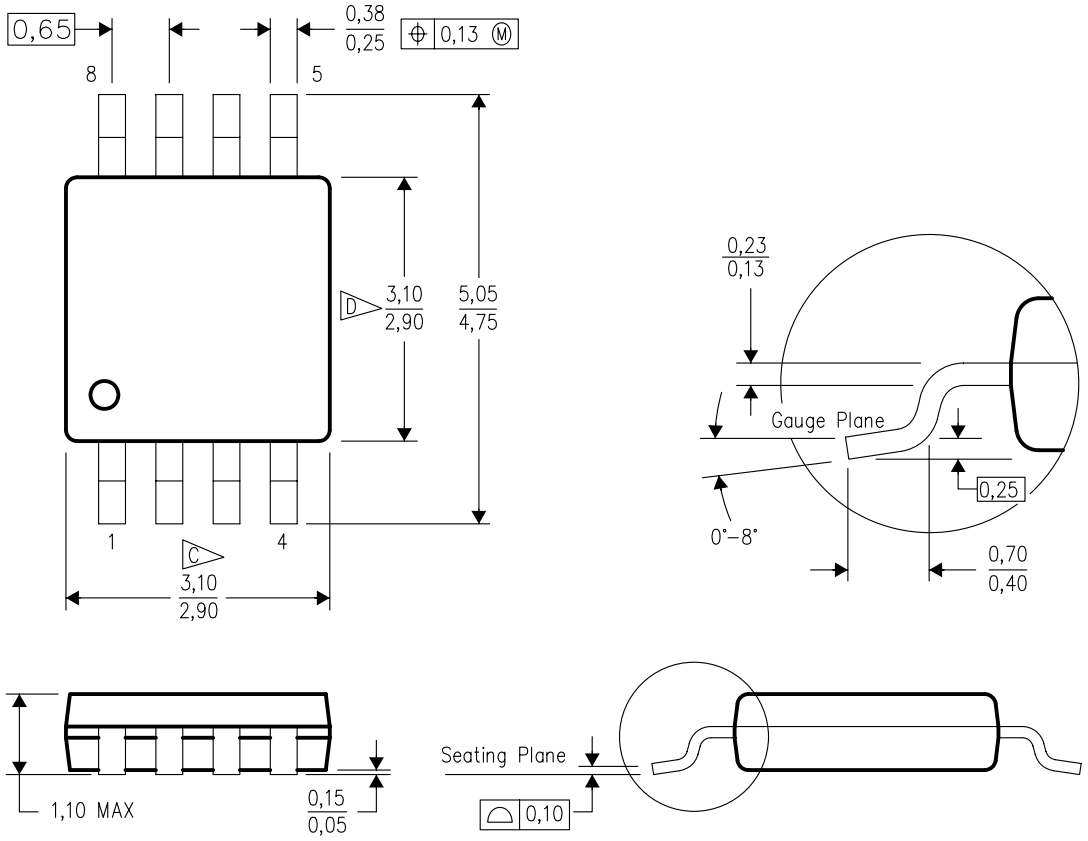
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

P (R–PDIP–T8) PLASTIC DUAL–IN–LINE PACKAGE



NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001 variation BA.

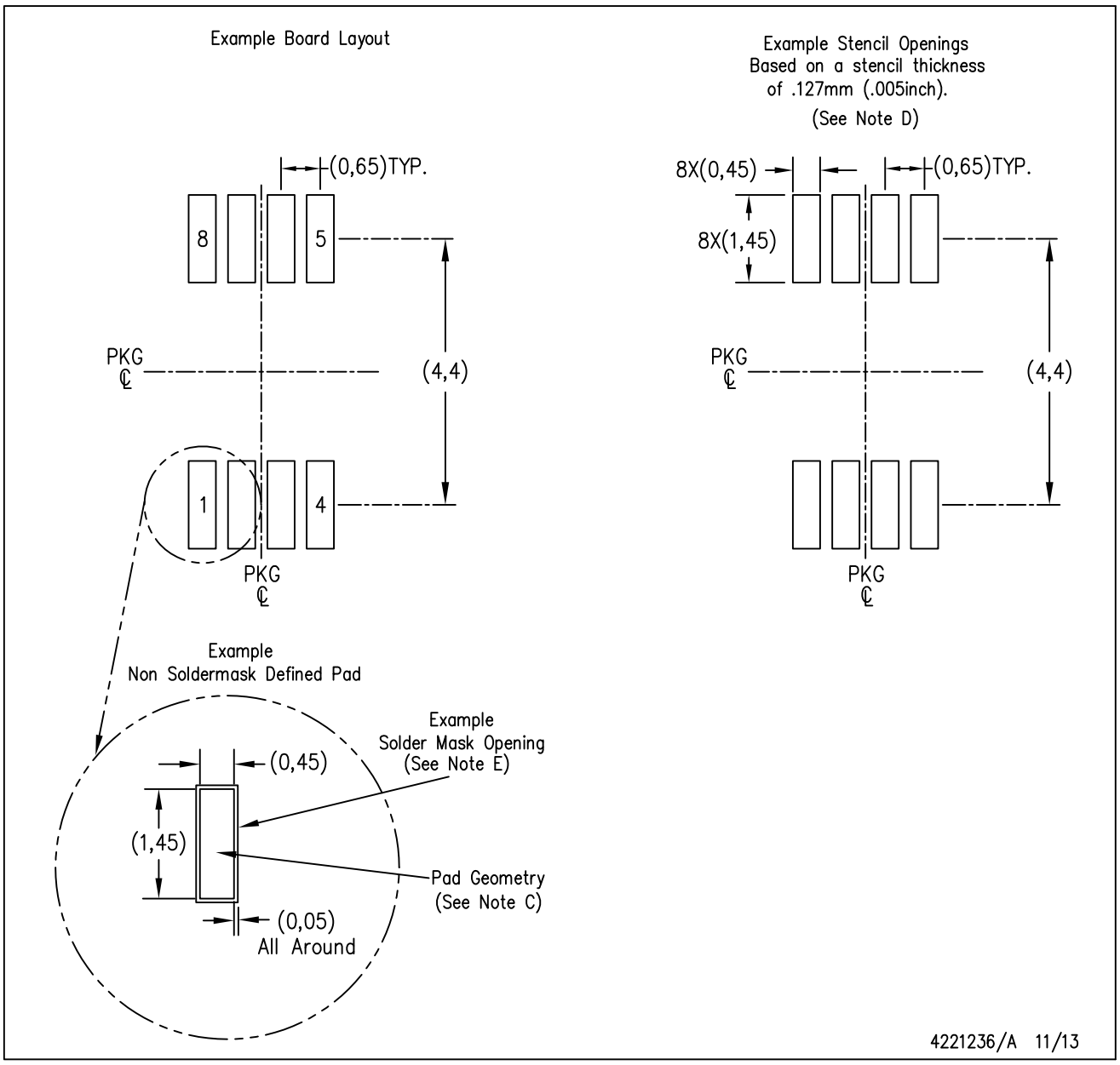
DGK (S–PDSO–G8) PLASTIC SMALL–OUTLINE PACKAGE



4073329/E 05/06

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 per end.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.50 per side.
E. Falls within JEDEC MO–187 variation AA, except interlead flash.

DGK (S–PDSO–G8) PLASTIC SMALL OUTLINE PACKAGE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

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